Less HIV Risk, More Syphilis: PrEP's Introduction and Risk Compensation

Savannah Ellison*

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Abstract

Pre-exposure prophylaxis (PrEP) is highly effective at preventing HIV. By lowering the expected cost of risky sex, PrEP access may lead users to reduce other risk-mitigating behaviors, potentially increasing the spread of other sexually transmitted infections like syphilis. However, more frequent screening of PrEP users could increase syphilis detection even if underlying incidence does not change. I estimate the effect of PrEP's 2012 introduction on syphilis using cross-state variation in the pre-treatment share of households consisting of male same-sex couples (MSS) as a proxy for PrEP access. Using Medicaid claims data, I construct three syphilis outcomes that together distinguish a change in infections from a change in detection. I estimate that greater PrEP access, proxied by a one-standard-deviation higher MSS, increases the syphilis diagnosis rate by 26.0 cases per 100,000 beneficiaries, an 81% increase relative to the 2011 mean. This implies that greater PrEP access increased observed syphilis diagnoses, but it cannot disentangle whether this was driven by an increase in syphilis in the population or by increased detection following changes in screening. The effect on the testing rate is about 7% of the 2011 mean and imprecise. Even the upper bound of the confidence interval is too small for increased testing to fully explain the rise in diagnoses. The yield rate, the share of positive syphilis tests, increases by about 59% relative to the 2011 mean. The large positive effect on the yield rate combined with the small, imprecise change in the testing rate suggests that a higher share of tests came back positive, indicating an increase in underlying syphilis incidence. Together, these results are consistent with PrEP access inducing risk compensation.

Keywords: PrEP, sexually transmitted infections, risk compensation, health policy

*Savannah Ellison: School of Economics, Georgia Institute of Technology, sellison8@gatech.edu. Thank you to Jason Lindo, Cici McNamara, Mayra Pineda-Torres, Richard Starr and members of the GA Tech HEPIC Brownbag and the Atlanta Athens Health Economists Group for valuable support and feedback.

1 Introduction

Pre-exposure prophylaxis for HIV (PrEP) medications are taken routinely to protect oneself from getting HIV. The first PrEP drug was approved by the U.S. Food and Drug Administration (FDA) in 2012 and is primarily intended for individuals at elevated risk of acquiring HIV, including men who have sex with men (MSM), people who inject drugs, and individuals whose sexual partners are living with HIV (CDC, 2021). PrEP has been cited as a major tool for ending the HIV epidemic because it is about 99% effective at preventing HIV acquisition through sex when taken as directed (HHS, 2019). Evidence since its introduction suggests PrEP has been effective at lowering HIV transmission at the population level as well, though its uptake is uneven across the United States (Beheshti et al., 2025; Sullivan et al., 2025).

While PrEP is effective at preventing HIV transmission, clinicians and researchers have been concerned that PrEP's availability may lower the perceived cost of risky sex and lead users to reduce other risk-mitigating behaviors like using condoms or limiting sexual partners. This could increase the incidence of other diseases associated with risky sex, such as bacterial sexually transmitted infections (STIs). This concern is especially salient given that bacterial STI rates in the United States had already been rising since the early 2000s and the same groups at risk of HIV tend to have higher incidence of bacterial STIs as well (CDC, 2024).

This is a classic example of risk compensation is, when a safety measure lowers the expected cost of a risky behavior, and individuals change their behavior based on how safe they perceive themselves to be. The concept was introduced by Peltzman (1975), who found that the introduction of significant automobile safety regulations did not decrease highway deaths and that pedestrians may have been made worse off, suggesting that people engaged in riskier driving in response to the new safety regulations. This concept has since been applied to a variety of settings and is particularly of interest in relation to sexual behavior and HIV treatment and prevention. Several papers study the effect of advances in HIV treatment, mainly the introduction of highly active antiretroviral therapy (HAART) on both HIV and syphilis incidence (Beheshti et al., 2026; Lakdawalla et al., 2006). In relation to HIV prevention, previous work studies the effect of syringe exchanges, aimed at preventing HIV transmission among people who inject drugs, on both HIV transmission and opioid overdoses (Packham, 2022).

Several descriptive studies have examined the association between PrEP use and risk compensation and reported increased rates of bacterial STIs and decreased condom use among PrEP users (Traeger et al., 2018). Small, causal studies have found mixed results in terms of risk compensation and PrEP. Some estimate increases in syphilis but argue that much of the increase is driven by increased STI screening of PrEP patients (Parker

et al., 2025; Quaife et al., 2020; von Schreeb et al., 2024). These studies provide important evidence on testing and diagnoses among individuals who initiate PrEP, but do not estimate the causal population-level effect of PrEP access on bacterial STI diagnoses and testing. At the United States population level, Eilam and Delhommer (2025) use CDC surveillance data and the 2000 Census share of male same-sex partnerships in a triple-difference design comparing men and women, and estimate that PrEP increased male syphilis rates by 16.2% between 2012 and 2018. Their syphilis measure includes only primary and secondary cases, whereas I include diagnoses at all stages. Like other studies, they cannot directly separate changes in incidence from changes in detection.

Observing changes in bacterial STI testing matters because, when PrEP was introduced, the Centers for Disease Control and Prevention (CDC) issued clinical guidelines directing providers to screen PrEP users for bacterial STIs like syphilis more frequently than if they were not taking PrEP (CDC, 2025). This means that an observed increase in the syphilis diagnosis rate could be impacted by two different mechanisms. It could represent a genuine increase in underlying syphilis in the population, or it could reflect more frequent testing catching infections that would otherwise have gone undiagnosed, or some combination of the two. As a result, existing studies can establish that diagnoses are rising but cannot say which channel is driving the increase.

In this paper, I estimate the effect of PrEP access on syphilis outcomes. Identifying the causal effect of PrEP on these outcomes is complicated because PrEP was approved nationally for prevention in 2012, leaving no clear comparison group of states without FDA-approved PrEP. I use a pre-treatment proxy to approximate state-level differences in PrEP access: the share of all households in a state consisting of cohabiting male same-sex couples (MSS), measured in the 2010 Census revised estimates. Men who have sex with men are the largest group for whom PrEP is recommended, so MSS proxies for the size of this group across states. To validate this proxy, I first estimate its relationship with the rate of PrEP patients per 100,000 Medicaid beneficiaries aged 15 and older in a given state-year. I then estimate the effect of differential PrEP access on the syphilis diagnosis rate, testing rate, and yield rate, defined as the number of tests classified as positive divided by the total number of tests performed. The diagnosis rate measures the level of observed syphilis cases and is useful to compare against other studies and surveillance statistics. The testing rate measures the level of syphilis screening occurring in a given state-year, which allows me to directly test for changes in detection. The yield rate measures the share of observed tests that come back positive, which I use to approximate underlying syphilis incidence while accounting for differences in testing volume. Estimating effects on all three outcomes helps parse out the extent to which changes in syphilis incidence and screening each affect any observed change

in the diagnosis rate.

I first show that MSS predicts PrEP uptake. I estimate that a one-standard-deviation higher MSS is associated with approximately 9.8 additional PrEP patients per 100,000 Medicaid beneficiaries after PrEP's introduction. The relationship is much stronger among men than among women, which is consistent with the goal of the male same-sex-couple share proxy.

In the main results, a one-standard-deviation higher MSS increases the syphilis diagnosis rate by 81.5% relative to its 2011 baseline mean. This implies that PrEP access increased diagnosed syphilis, but a rise in diagnoses alone cannot distinguish more infections from more detection. The estimated effect on the testing rate is small and imprecise. This suggests that increased screening is not driving the increase in observed diagnoses. The estimated effect on the yield rate is large, a 59.3% increase relative to its baseline mean, and is significant at the 10% level. The coefficient suggests that a higher share of tests came back positive, consistent with more infections among those tested. Interpreting all three results together suggests an increase in underlying syphilis incidence in states with higher PrEP access, which is consistent with PrEP inducing risk compensation. However, the imprecision of the testing and yield rates prevents me from ruling out part of the increase in the diagnosis rate being driven by testing.

The estimated effects are concentrated among men, consistent with their substantially greater use of PrEP. Among men, a one-standard-deviation higher MSS increases the diagnosis rate by 39.5 cases per 100,000, a 111.0% increase relative to the male baseline mean, an estimate that is statistically significant at the 5% level. The male testing estimate is larger in proportional terms than the pooled estimate, and male testing rises in the later years of the sample, consistent with the more frequent screening recommended for PrEP users. Its confidence interval, however, again rules out increases large enough to account for the rise in male diagnoses. The male yield estimate is positive but imprecise. Among women, the estimates are positive but smaller and not statistically significant.

The results hold up in several additional checks. Event-study estimates show no evidence of a differential change between 2010 and 2011, the two years observed before PrEP's introduction, and the diagnosis and yield effects emerge after 2014, in step with the delayed rise in PrEP uptake. The main results are nearly unchanged when Kansas, Maine, Idaho, and Mississippi, which have incomplete observations in the claims data, are excluded. Finally, leave-one-out estimates show that the diagnosis, yield, and testing point estimates remain positive when any single state is excluded. However, the diagnosis and yield estimates are smaller when Delaware or Nevada is omitted.

2 Background

2.1 HIV and PrEP

Human Immunodeficiency Virus (HIV) emerged in the United States in the 1980s. Doctors and researchers were initially unaware of how the virus spread or how to treat it, making it very deadly. Although infections were initially concentrated among men who have sex with men, by the mid-1990s HIV was the leading cause of death for men aged 25-44 and one of the leading causes of death for women aged 25-44 (CDC, 1996). This changed with the introduction of Highly Active Antiretroviral Therapy (HAART) in 1996, a combination of medications that suppressed viral replication, which decreased the mortality of the disease and made it a more manageable chronic condition. In the years that followed, practitioners began using antiretroviral medications to prevent HIV transmission to HIV-negative individuals. First, clinicians began using post-exposure prophylaxis (PEP), a 28-day course of antiretroviral medications taken after a potential exposure to HIV. In 2005, the CDC issued recommendations extending PEP to non-occupational exposures (CDC, 2005). Then, pre-exposure prophylaxis for HIV (PrEP) was introduced. Unlike PEP, which is taken after a potential exposure, PrEP is taken consistently before any known exposure by people at high risk of acquiring HIV. PrEP was approved by the FDA in July 2012 (U.S. Food and Drug Administration, 2012). When taken as directed, PrEP is approximately 99% effective at preventing HIV acquisition through sex (CDC, 2021). Access to knowledgeable prescribers, insurance coverage, and consistent daily adherence all affect PrEP's true impact outside of a clinical trial setting (Cambou & Landovitz, 2021). Previous work has estimated that areas with the highest PrEP uptake saw a significant decrease in HIV diagnoses, but PrEP uptake was concentrated in areas with larger White populations (Beheshti et al., 2025).

2.2 Bacterial STIs

Bacterial sexually transmitted infections (STIs) are infections transmitted through sexual contact and include chlamydia, gonorrhea, and syphilis, the three most commonly reported bacterial STIs in the United States. These infections are frequently asymptomatic or have mild symptoms that can be missed. This means that without consistent screening, many cases may be undiagnosed. However, some cases of bacterial STIs can have severe complications including infertility, heart problems, and death (CDC, 2026a).

I focus on syphilis and include diagnoses at all stages of infection. Early-stage diagnoses typically reflect recent transmission, while later-stage and latent diagnoses may reflect infections acquired substantially earlier. The diagnosis measure therefore captures both recent

infections and previously acquired infections that are detected during the sample period. Syphilis is an ideal bacterial STI to focus on in this context for a few reasons. First, men who have sex with men already had disproportionate rates of syphilis before PrEP's introduction (Beheshti et al., 2026). Finally, syphilis is particularly relevant in the context of HIV prevention, because having an untreated syphilis infection increases the likelihood of someone contracting HIV (CDC, 2012).

Screening recommendations vary substantially based on a variety of risk factors including sex, age, pregnancy, and self-reported sexual behavior and drug use. The CDC recommends annual chlamydia and gonorrhea screening for sexually active women under age 25 and for older women at increased risk (CDC, 2026c; U.S. Preventive Services Task Force, 2021). Syphilis screening is recommended during pregnancy and for other populations at elevated risk. Sexually active men who have sex with men are recommended to undergo syphilis, gonorrhea, and chlamydia screening annually, with screening every three to six months recommended for those at increased risk, including people using PrEP (CDC, 2026b).

Although bacterial STIs and HIV are both sexually transmitted, they differ in important ways. HIV is caused by a virus and can be treated but not cured. Bacterial STIs are caused by bacteria and are generally curable with antibiotics. Thus, while PrEP prevents the acquisition of HIV, it does not prevent bacterial STIs. There is currently no comparable medication routinely used as pre-exposure prophylaxis for bacterial STIs.

3 Data

The main dataset I use in this analysis is individual-level administrative claims data from the Medicaid Analytic eXtract (MAX) and Transformed Medicaid Statistical Information System Analytic Files (TAF) from 2010-2022. These data collectively cover Medicaid beneficiaries across all 50 states. All individual-level claims for Kansas and Maine are missing in 2010 because of data-reporting issues (Mathematica Policy Research, 2014). Additionally, all individual-level outpatient claims for Idaho are missing in 2011 (Mathematica Policy Research, 2017). In addition, Mississippi's 2021 claims cannot be reliably linked to beneficiaries, which CMS rates as unusable for that year (**cms˙dq˙atlas**). I include these four states in the main specification and show that the results are robust to their exclusion in Section 6.3. Individual level claims data are well suited to this analysis because they capture both diagnoses and tests, allowing me to estimate effects separately on diagnosis, testing, and yield rates. I supplement the claims data with the 2010 Decennial Census, which I use to construct the male same-sex couple share, as well as demographic controls constructed from the American Community Survey (ACS) Public Use Microdata Sample (PUMS) and

an indicator for Medicaid expansion.

Using Medicaid data means I am analyzing a specific subset of the United States population. Medicaid enrollees are by construction lower-income and are also disproportionately younger and more racially diverse than the United States population as a whole (KFF, 2026). This is a key population for both HIV and bacterial STI risk. Young people disproportionately contract bacterial STIs. In 2023, people aged 15-24 accounted for nearly half (48.2%) of reported chlamydia, gonorrhea, and syphilis cases nationally, despite comprising only about 13% of the U.S. population (CDC, 2024). Black Americans are also disproportionately affected by both HIV and bacterial STIs. In 2023, Black/African American people accounted for approximately 38% of new HIV diagnoses and 32.4% of chlamydia, gonorrhea, and syphilis cases despite representing only about 12% of the U.S. population (CDC, 2024).

3.1 PrEP Identification

I use the prescription drug (Rx) files to identify PrEP prescriptions, which I use to estimate the quasi-first-stage relationship between MSS and PrEP use. PrEP consists of a small number of branded and generic drugs. Truvada (emtricitabine/tenofovir disoproxil fumarate) was the first drug approved for PrEP use. Descovy (emtricitabine/tenofovir alafenamide) was approved for PrEP in 2019. Generic Truvada became available in 2020, and Apretude, an injectable option, became available in 2022. A handful of other PrEP drugs entered the market after 2022 and are therefore not included in this analysis.

Truvada and Descovy are approved for both HIV treatment as part of antiretroviral therapy and HIV prevention, PEP and PrEP. This creates a measurement obstacle, because the same national drug code (NDC) can appear for either use, and no ICD code distinguished PrEP prescriptions during the sample period (CDC, 2021; HRSA, 2024). I exploit the clinical fact that when Truvada or Descovy is prescribed as part of HIV treatment, it is always co-prescribed with at least one additional antiretroviral drug, while PrEP is prescribed as a standalone regimen (NIH, 2021). I classify a Truvada or Descovy fill as a PrEP prescription if it is not accompanied by a co-prescribed antiretroviral therapy drug within 90 days of the fill. Because PEP is a single 28-day course, I additionally require a beneficiary's cumulative Truvada or Descovy fills to exceed 30 days' supply before classifying them as a PrEP user, so that standalone PEP courses are not counted as PrEP. Apretude is only prescribed for PrEP, which makes it straightforward to identify using NDC codes or HCPCS code J0739. The only caveat is that Apretude must be pulled from both prescription and outpatient files because of state-by-state variation in the billing of injectable drugs. The NDC codes used to identify PrEP and co-prescribed antiretroviral fills are listed in Appendix B.

Figure 1 shows the growth in PrEP patients among Medicaid beneficiaries aged 15 and older over the study period by sex. PrEP use is low in the first few years after FDA approval, then starts growing around 2014. This pattern is useful for interpreting the event-study estimates later in the paper where a similar delayed effect is reflected. This delayed uptake trend is also present in the all-payer trend of PrEP uptake in the United States, which I compare with the Medicaid trend in Section 3.2.

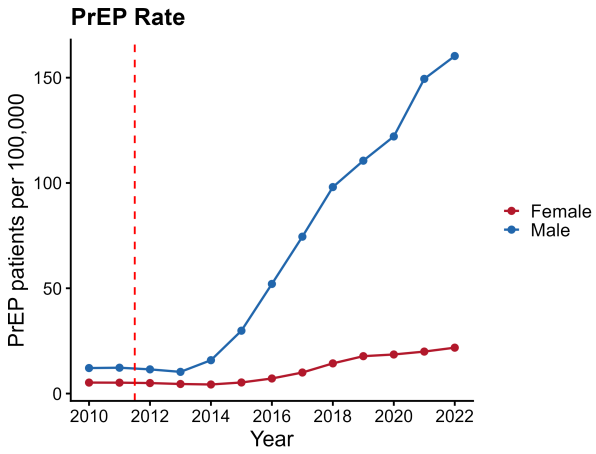


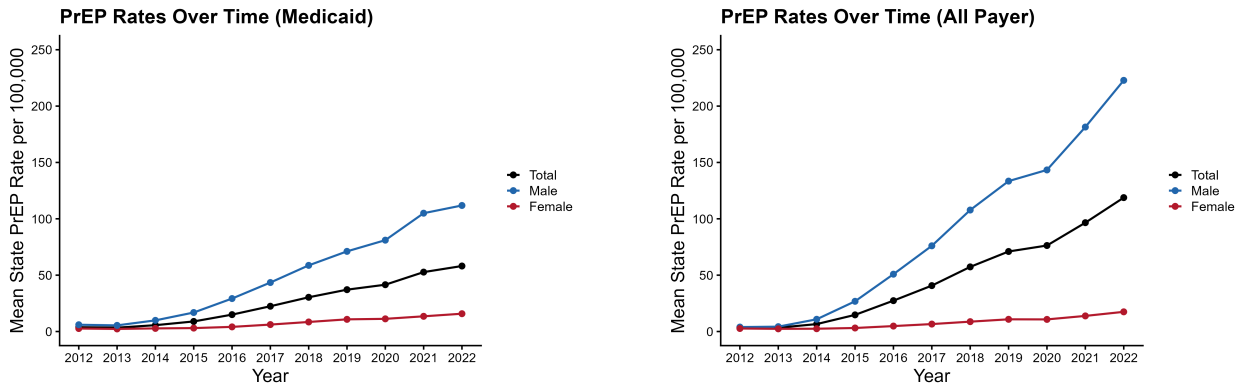
Figure 1: PrEP Rate by Sex, Medicaid Beneficiaries Aged 15 and Older

The PrEP rate is the number of patients with a PrEP prescription per 100,000 male or female Medicaid beneficiaries aged 15 and older, measured annually. Rates are computed from counts summed across the 50 states. The dashed vertical line marks PrEP's 2012 FDA approval.

The graph also shows that the majority of PrEP users are men. This motivates heterogeneity analysis by sex later in the paper.

3.2 Medicaid and All-Payer PrEP Rates

Because sexually transmitted infections are spread through communities and I argue that the male same-sex couple share is a measure of PrEP access, it is important to know how all-payer PrEP use compares to the PrEP use among Medicaid beneficiaries. To check this, I use all-payer PrEP rates from AIDSVu, which reports the number of PrEP users in each state-year using IQVIA prescription data (AIDSVu, 2026). Figure 2 compares PrEP rates among Medicaid beneficiaries with all-payer PrEP rates. Both series are unweighted averages across the 50 states. The two graphs show similar trends. PrEP use is near zero in 2012 and 2013, then it begins rising around 2014 and grows steadily through 2022. Men also account for the majority of PrEP users in both the Medicaid and all-payer data. The rate of PrEP use is higher in the all-payer data. This is unsurprising because others have documented that the majority of PrEP users are not covered by Medicaid (Huang et al., 2018).



(a) Medicaid

(b) All Payer

Figure 2: PrEP Rates Over Time: Medicaid vs. All Payer

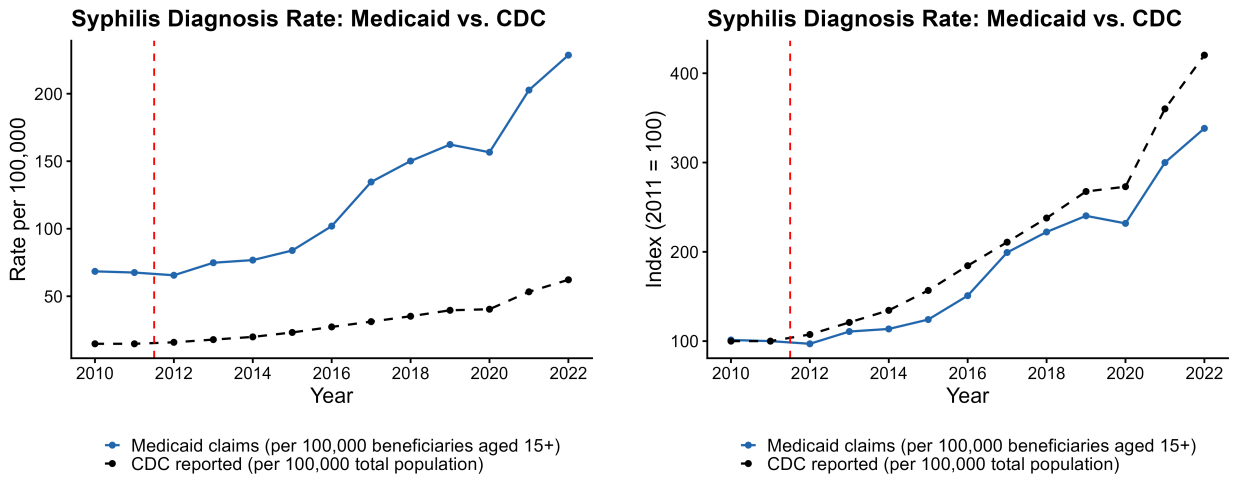
Panel (a) shows the average across the 50 states of the Medicaid PrEP rate, defined as PrEP patients per 100,000 Medicaid beneficiaries aged 15 and older. Panel (b) shows the average across the 50 states of the all-payer PrEP rate from AIDSVu, defined as PrEP users per 100,000 residents aged 13 and older. AIDSVu identifies PrEP users from IQVIA prescription data, using an algorithm that excludes prescriptions for HIV treatment, chronic hepatitis B, and post-exposure prophylaxis, and adjusts for prescriptions not captured in the data. DC is excluded from both series.

3.3 STI Outcomes

From the individual-level claims, I construct three state-year outcomes: the syphilis diagnosis rate, testing rate, and yield rate. I use CPT codes to identify syphilis tests and ICD-9 and ICD-10 diagnosis codes to identify syphilis diagnoses at all stages. The testing rate includes all observed syphilis tests. I classify a syphilis test as positive if the same beneficiary has a claim containing a syphilis diagnosis code within 30 days following the observed test. The diagnosis and testing rates are expressed per 100,000 Medicaid beneficiaries aged 15 and older in each state-year. A beneficiary can contribute more than one positive diagnosis in a year if the positive tests are more than 90 days apart. The diagnosis rate is the number of unique positive diagnosis events per 100,000 beneficiaries. The yield rate is the number of positive testing events divided by the total number of syphilis test events, multiplied by 100. The diagnosis and procedure codes used to construct these outcomes are listed in Appendix B. Following CMS guidelines, I suppress state-year-sex counts of each outcome that pertain to between 1 and 10 patients. I set suppressed counts to zero in my analysis.

Figure 3 compares the aggregate syphilis diagnosis rate in the Medicaid claims with the national total syphilis rate reported by the CDC, which covers all stages of syphilis, including congenital syphilis, and rose from 14.8 cases per 100,000 population in 2011 to 62.2 in 2022 (CDC, 2024). Levels are higher in the claims, reflecting the restriction to Medicaid beneficiaries aged 15 and older, a younger and lower-income population than the U.S. as a whole, and a case definition that allows a beneficiary to contribute more than one case per year. Indexed to 2011, however, the two series follow similar trajectories, remaining

flat through 2012 and rising steadily thereafter, with a sharp increase in 2021-2022. The Medicaid series also declines in 2020, consistent with disruptions to routine care and testing during the COVID-19 pandemic. This similarity provides reassurance that the diagnosis measure captures the broader resurgence in syphilis documented in surveillance data.



(a) Rate per 100,000

(b) Indexed, 2011 = 100

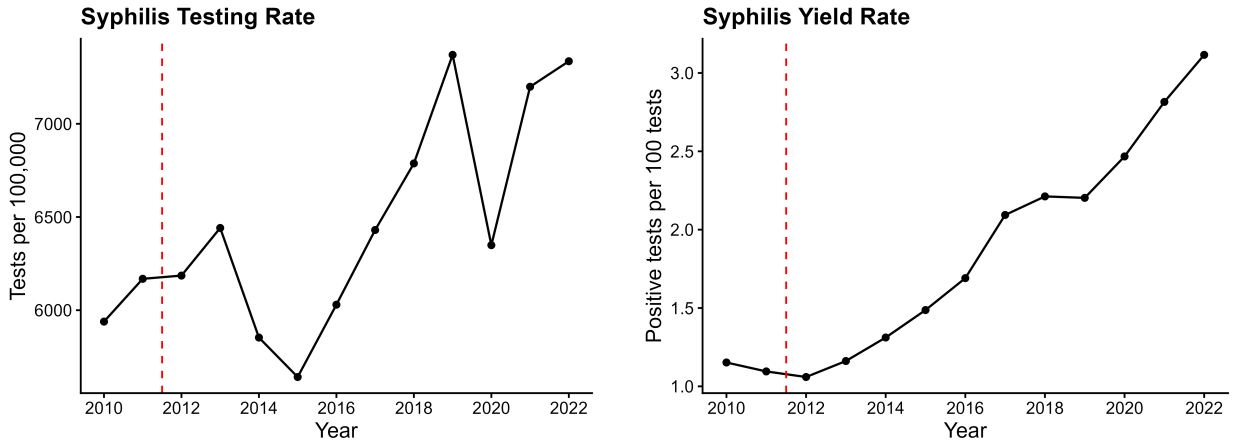
Figure 3: Syphilis Diagnosis Rate: Medicaid Claims and CDC Surveillance

The Medicaid series is the total number of syphilis cases divided by the total number of Medicaid beneficiaries aged 15 and older across the 50 states, per 100,000. The CDC series is reported total syphilis (all stages, congenital syphilis, and stage not stated) per 100,000 U.S. population, from CDC STI Surveillance, Table 1. The dashed vertical line marks PrEP's FDA approval in 2012.

Rising diagnoses alone are difficult to interpret, however, since they could reflect greater underlying transmission, more intensive screening, or both. Figure 4 shows national trends in the syphilis testing and yield rates. The yield rate scales cases by the total number of tests performed, providing a measure that is less mechanically tied to changes in testing volume. Nationally, the testing rate rises only modestly and fluctuates from year to year, while the yield rate roughly triples between 2012 and 2022.

3.4 Male Same-Sex Couples Proxy for PrEP Access

Identifying the causal effect of PrEP's introduction is complicated by the lack of a clean control group, because all states get access at the same time. To address this, I use the U.S. Census Bureau's 2010 Decennial Census revised estimates of the share of male same-sex couple households among all households in a state as a proxy, fixed before PrEP's approval, for the size of each state's population likely to take up PrEP. The Census Bureau released revised same-sex couple household counts after identifying a coding error that inflated the original 2010 estimates. I use the corrected counts, described in O'Connell and Feliz (2011).



(a) Testing Rate

(b) Yield Rate

Figure 4: National Syphilis Testing and Yield Rates, Medicaid Beneficiaries

The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Rates are computed from counts summed across the 50 states. The dashed vertical line marks PrEP's FDA approval in 2012.

The average state male same-sex couple share is 0.247 percent, with a standard deviation of 0.0878 percentage points. This proxy is imperfect because it does not capture single gay men, people who inject drugs, or women who use PrEP. That said, the measure is highly predictive of subsequent state-level PrEP use.

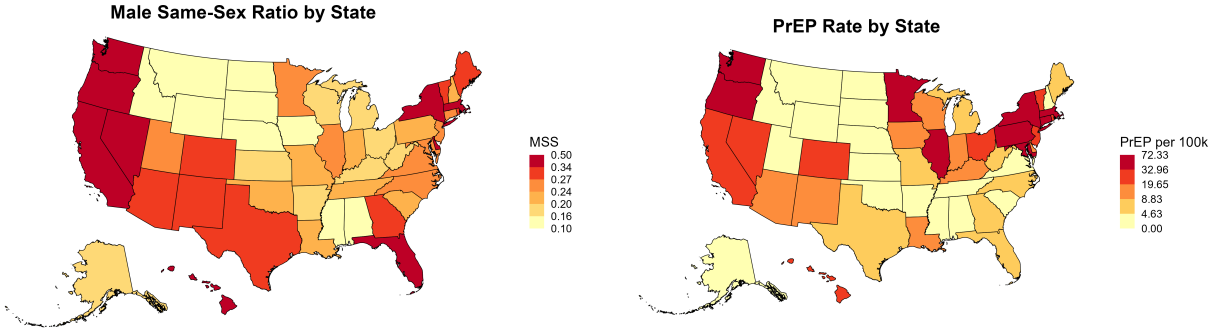


Figure 5: MSS and the State PrEP Rate

The left map shows each state's 2010 Census male same-sex couple share (MSS), in percent. The right map shows each state's 2016 PrEP rate, defined as PrEP patients per 100,000 Medicaid beneficiaries aged 15 and older. In each map, states are shaded by sextile of the 50-state distribution. DC is excluded.

Figure 5 maps the male same-sex couple share and the 2016 state Medicaid PrEP rate. It shows that states with high MSS tend to have high Medicaid PrEP rates as well. Oregon, Washington, New York, and Massachusetts are all in the highest sextile of both MSS and PrEP use. Similarly, states like North Dakota, South Dakota, Montana, and Idaho are in

the lowest quantile for both MSS and PrEP use. Of course this relationship is not perfect. Florida, Nevada, and Texas have high male same-sex couple shares, but have moderate PrEP rates. Conversely, Illinois and Minnesota have high PrEP rates relative to their MSS. To formally validate my use of MSS as a proxy for PrEP access, I estimate the effect of MSS on the Medicaid PrEP rate in the results section.

3.5 Controls

My preferred empirical specifications use time-varying state-level demographic controls to avoid confounding the effect of PrEP access on bacterial STI outcomes. I control for the demographic composition of each state's Medicaid population using the ACS 1-year PUMS (U.S. Census Bureau, 2026). I define Medicaid enrollees as civilian noninstitutionalized people aged 0-64 covered by Medicaid, medical assistance, or another government assistance plan for people with low incomes or a disability (the ACS HINS4 item), and compute each share with the person weights. In the pooled specifications, the controls are the shares of Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic, the share of enrollees aged 0-64 who are adults aged 19-64, and the male share of enrollees aged 19-64. In the sex-specific and triple-difference specifications, I compute the same race, ethnicity, and age shares among enrollees of the same sex, so the controls differ for men and women in the same state-year, and I omit the male share. The Census Bureau did not release standard ACS 1-year estimates for 2020, so I linearly interpolate 2020 values within each state and sex. I also treat shares with a relative standard error above 30%, estimated with the 80 replicate weights, as missing and fill them by interpolation within state and sex. These are mainly the Black share in a few small states.

Table 1 displays summary statistics for the data in my sample split by states below and above the median MSS and for the years before and after PrEP's approval. States with above-median male same-sex couple shares expanded adult Medicaid at a higher rate than below-median MSS states. Above-median MSS states also have Medicaid populations with larger Hispanic and adult shares. In both groups of states PrEP use and syphilis diagnosis, testing, and yield rates rose after 2012. The increases appear larger for all outcomes in the above-median group. However, these descriptive statistics do not account for differences in Medicaid expansion and other differences between above and below median PrEP access states that may affect the outcomes, but are controlled for in the regression estimates using demographic controls and state fixed effects.

	Below-median MSS		Above-median MSS	
	2010–2011	2012–2022	2010–2011	2012–2022
A. PrEP exposure				
MSS (%)	0.177 (0.038)	0.177 (0.037)	0.318 (0.062)	0.318 (0.062)
B. PrEP use				
PrEP rate, pooled	1.3 (2.2)	13.7 (21.3)	6.5 (6.3)	37.0 (37.8)
PrEP rate, male	2.3 (4.6)	26.4 (40.8)	10.2 (9.7)	71.2 (72.2)
PrEP rate, female	0.8 (1.6)	4.3 (7.8)	4.6 (5.4)	10.4 (12.0)
C. Syphilis outcomes, pooled				
Diagnosis rate	25.5 (39.2)	77.0 (90.1)	36.7 (37.7)	126.3 (135.5)
Testing rate	4,128 (2,730)	4,716 (2,232)	4,354 (2,989)	5,677 (3,039)
Yield rate	0.48 (0.57)	1.40 (1.17)	0.68 (0.50)	2.05 (2.03)
D. Syphilis outcomes, male				
Diagnosis rate	23.5 (31.6)	82.9 (102.9)	44.5 (47.0)	166.7 (178.1)
Testing rate	1,399 (941)	2,080 (1,369)	2,141 (1,918)	3,392 (2,251)
Yield rate	1.46 (1.90)	3.23 (2.60)	1.75 (1.30)	4.39 (3.38)
E. Controls (share of Medicaid enrollees)				
White NH share of Medicaid enrollees 0–64 (%)	60.5 (13.9)	59.2 (13.7)	43.3 (19.0)	42.5 (18.5)
Black NH share of Medicaid enrollees 0–64 (%)	19.5 (17.1)	18.3 (15.9)	17.5 (14.7)	16.6 (13.3)
Hispanic share of Medicaid enrollees 0–64 (%)	9.6 (5.8)	10.6 (6.0)	28.1 (16.8)	28.6 (16.2)
Adult (19–64) share of Medicaid enrollees 0–64 (%)	37.9 (5.5)	44.5 (7.6)	43.5 (8.2)	49.7 (8.7)
Male share of Medicaid enrollees 19–64 (%)	38.8 (2.9)	40.8 (2.9)	39.8 (2.6)	42.2 (2.6)
Medicaid expansion in effect (%)	0.0 (0.0)	43.6 (49.7)	0.0 (0.0)	62.9 (48.4)
F. Sample				
Beneficiaries aged 15+ (thousands)	533 (456)	713 (641)	1,227 (1,787)	1,660 (2,363)
States	25	25	25	25
State-years	48	275	49	275

Table 1: Summary Statistics

Each cell reports the unweighted mean across state-years, with the standard deviation in parentheses. States are divided at the median 2010 male same-sex couple share (MSS) across the 50 states. The PrEP, diagnosis, and testing rates are per 100,000 Medicaid beneficiaries aged 15 and older (male or female beneficiaries for the sex-specific rows). The yield rate is syphilis cases per 100 syphilis tests. Control shares are percentages of each state's Medicaid enrollees, from ACS 1-year PUMS. DC is excluded. Kansas and Maine are missing in 2010, Idaho's outpatient claims are missing in 2011, and Mississippi's 2021 claims cannot be linked to beneficiaries. State and state-year counts refer to the pooled syphilis sample.

4 Empirical Strategy

I estimate the effect of differential PrEP access on bacterial STI outcomes by exploiting cross-state variation in the size of the population likely to take up PrEP following its FDA approval in 2012. I consider three syphilis outcomes: the diagnosis rate, testing rate, and yield rate. As described in Section 3, I use the pre-period share of male same-sex couple households in a state, measured in the 2010 Census, as a proxy for differential PrEP access and estimate a continuous-treatment difference-in-differences specification that allows post-2012 changes in outcomes to vary with the state's pre-period MSS share:

$$Y_{st} = \beta \left(\text{MSS}_s \times \text{Post}_t \right) + \gamma X_{st} + \mu_s + \tau_t + \epsilon_{st} \tag{1}$$

where MSS_s is the 2010 Census male same-sex couple share in state s, expressed in percentage points, and Post_t is an indicator equal to one in 2012 and subsequent years. The coefficient β captures the differential post-2012 change in each outcome caused by a one-percentage-point higher pre-period male same-sex couple share (MSS). To make the results easier to interpret

relative to the distribution of MSS, I report effects scaled to a one-standard-deviation (0.0878-percentage-point) difference in MSS. Standard errors are clustered at the state level because treatment intensity varies at the state level.

In addition to state and year fixed effects, a subset of specifications includes the time-varying, state-level controls X_{st} described in Section 3. These controls include the White, Black, and Hispanic shares of Medicaid enrollees, the adult share of enrollees, the male share of adult enrollees, and an indicator for Medicaid expansion. In the sex-specific specifications, the shares are computed among enrollees of the same sex and the male share is omitted.

Quasi-First Stage. To validate the proxy, I estimate the relationship between the male same-sex couple share and the PrEP rate:

$$\text{PrEP}_{st} = \beta \left(\text{MSS}_s \times \text{Post}_t\right) + \gamma X_{st} + \mu_s + \tau_t + \epsilon_{st} \tag{2}$$

where PrEP_{st} is the rate of patients prescribed PrEP per 100,000 Medicaid beneficiaries aged 15 and older in state s in year t. A positive and significant $\hat{\beta}$ indicates that states with higher male same-sex couple shares experienced differential PrEP uptake after 2012, supporting the proxy.

Event Study. The identifying assumption behind equation (1) is that, absent the introduction of PrEP, post-2012 changes in STI outcomes would not have been systematically related to the state's pre-period MSS share. This assumption would be violated if states with more male same-sex couples were on differential STI trends prior to 2012 for reasons unrelated to PrEP. To assess this, I estimate an event-study version of equation (1), replacing the single $\text{MSS}_s \times \text{Post}_t$ interaction with a full set of year-by-treatment-intensity interactions:

$$Y_{st} = \sum_{\substack{k=2010 \\ k \neq 2011}}^{2022} \beta_k \left(\text{MSS}_s \times \mathbf{1}[t = k]\right) + \gamma X_{st} + \mu_s + \tau_t + \epsilon_{st} \tag{3}$$

where $\mathbf{1}[t = k]$ is an indicator for year k, and 2011, the year immediately preceding FDA approval, is the omitted reference category. The coefficient β_{2010} captures the differential change between 2010 and 2011 associated with the state's pre-period MSS share. An estimate close to zero provides evidence against differential changes in STI outcomes immediately prior to PrEP's introduction. The coefficients $\{\beta_k\}_{k \geq 2012}$ trace out the dynamic pattern of differential post-2012 changes in outcomes. I estimate equation (3) separately for each outcome, and apply the same specification to equation (2) to assess pre-trends in the quasi-first stage. Because only one pre-period coefficient is available, a near-zero 2010 estimate is limited evidence for parallel trends. Section 6.1 therefore reports bounds on the post-period effects that allow for violations of parallel trends, following Rambachan and Roth (2023).

To estimate whether the male same-sex couple share predicts PrEP uptake differentially by sex, I estimate equation (1) separately for the male and female subsamples. Because PrEP uptake is substantially greater among men, we would expect to see a larger effect for men. However, the effect among women need not be zero. Some women are at high risk of contracting HIV and thus are the target population to take PrEP. I therefore interpret the sex-specific estimates as evidence on whether the relative magnitude of the effects is consistent with differential PrEP use across sex.

The separately estimated male and female specifications do not formally test whether the effects differ between men and women. To test this directly, I stack the male and female state-year panels and estimate a triple-difference specification:

$$Y_{sgt} = \delta \left(\text{MSS}_s \times \text{Post}_t \times \text{Male}_g\right) + \beta \left(\text{MSS}_s \times \text{Post}_t\right) + \gamma X_{sgt} + \mu_{sg} + \tau_{gt} + \epsilon_{sgt} \qquad (4)$$

where g indexes sex, Male_g is an indicator for male beneficiaries, X_{sgt} are the sex-specific demographic controls and the Medicaid expansion indicator, μ_{sg} are state-by-sex fixed effects, and τ_{gt} are sex-by-year fixed effects. The state-by-sex and sex-by-year fixed effects absorb the lower-order $\text{MSS}_s \times \text{Male}_g$ and $\text{Post}_t \times \text{Male}_g$ terms. The coefficient δ measures the differential post-2012 change in each outcome among men relative to women associated with a one-percentage-point higher MSS. The identifying assumption is that, absent PrEP, the gap between male and female outcomes would have changed similarly in states with higher and lower MSS. Shocks that affect men and women in a state equally, such as changes in state Medicaid policy, do not violate this assumption. Standard errors are clustered at the state level.

Because women also use PrEP and may be affected through sexual networks with men, δ is not the full effect of PrEP access on men. It is the difference between the male and female effects, and it understates the male effect to the extent that women's outcomes also respond. A positive δ for the PrEP rate confirms that the proxy captures differential PrEP uptake among men, and a positive δ for the syphilis outcomes indicates that the effects are concentrated where PrEP access is greatest. This design is similar to the triple-difference comparison of men and women in Eilam and Delhommer (2025). Section 6.2 reports the estimates.

I assess the validity and sensitivity of the main results in several additional ways in Section 6. First, I construct confidence intervals that remain valid under bounded violations of parallel trends. Second, I estimate the triple-difference specification comparing men and women. Third, I re-estimate the specifications excluding Kansas, Maine, Idaho, and Missis-

sippi, which have incomplete or unlinkable claims data during part of the sample period, as described in Section 3. Fourth, I conduct a leave-one-out analysis, re-estimating the model after excluding each state one by one to assess whether results are driven by any single state. All results are estimated over the 2010-2022 sample period.

5 Results

In this section, I first estimate the quasi-first-stage effect of the male same-sex share on the PrEP rate. This demonstrates whether my chosen proxy predicts PrEP uptake following PrEP's introduction. Then, I move on to the main results where I estimate the effects of differential PrEP access on the syphilis diagnosis rate, which conveys the effect on observed syphilis diagnoses, the testing rate, which conveys the effect on syphilis screening, and on the syphilis yield rate, which I use to approximate changes in underlying syphilis incidence by accounting for differences in testing volume. Section 5.3 reports estimates for male and female beneficiaries separately. Section 6 reports sensitivity to violations of parallel trends, triple-difference estimates, estimates excluding states with incomplete claims, and leave-one-out estimates.

5.1 Effect of Male Same-Sex Couple Share on the PrEP Rate

This quasi-first-stage provides a direct check that the MSS measure captures meaningful cross-state variation in PrEP access. Figure 6 presents the event-study estimates for the PrEP rate. The 2010 coefficient is close to zero and not statistically significant relative to the omitted 2011 year, providing no evidence of differential PrEP trends immediately prior to FDA approval. The coefficients remain near zero in the first three years following PrEP's FDA approval, then rise steadily. This pattern mirrors the raw trend of PrEP take-up shown in Figure 1. The event study supports the use of the pre-treatment MSS share as a proxy for differential PrEP access across states.

Table 2 shows a positive and statistically significant relationship between MSS and subsequent PrEP uptake. In the preferred specification with demographic controls, the coefficient on MSS $\times$ Post is 112.1 ($p < 0.01$), compared with 129.7 without controls. A one-standard-deviation higher MSS (0.0878 percentage points) is associated with approximately 9.8 additional PrEP patients per 100,000 Medicaid beneficiaries aged 15 and older following PrEP's introduction, more than twice the 2011 baseline mean of 4.0.

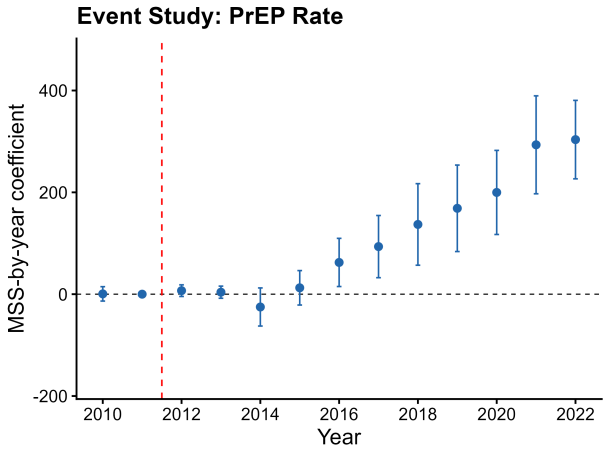


Figure 6: Event Study: PrEP Rate

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3) applied to the quasi-first stage in Equation (2), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls. The dashed vertical line marks PrEP's 2012 FDA approval.

Dependent Variable:	PrEP Rate	
Model:	(1)	(2)
Variables		
MSS × Post	129.7***	112.1***
	(20.96)	(19.39)
Demographic Controls	No	Yes
2011 Baseline Mean	4.03	4.03
Fixed-effects		
State	Yes	Yes
Year	Yes	Yes
Fit statistics		
Observations	648	648

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 2: Quasi-First Stage: Effect of MSS × Post on the PrEP Rate

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic, the share of enrollees aged 0-64 who are adults aged 19-64, and the male share of enrollees aged 19-64. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

5.2 Syphilis

Figure 7 presents the event-study estimates for the diagnosis rate. The 2010 coefficient is close to zero and not statistically significant relative to the omitted 2011 reference year, and the coefficients remain near zero through 2014, mirroring the delayed rise in PrEP uptake in Figure 6. From 2015 onward the coefficients are positive and rise through 2019. Individual post-period coefficients are imprecisely estimated, however, and most 95% confidence

intervals include zero.

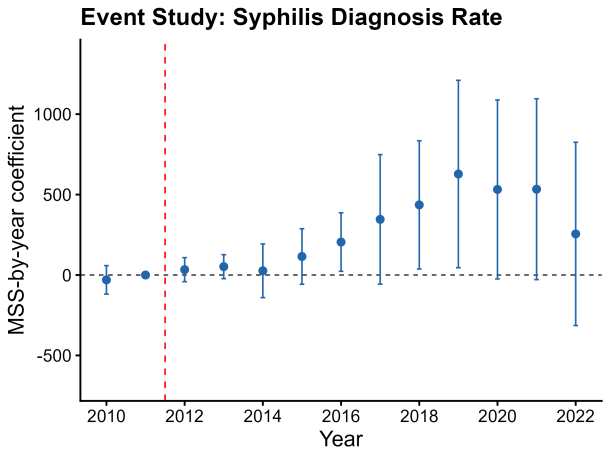


Figure 7: Event Study: Syphilis Diagnosis Rate

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls. The dashed vertical line marks PrEP's 2012 FDA approval.

Figure 8 presents the corresponding estimates for the testing and yield rates. The testing-rate coefficients are noisy and do not exhibit a clear post-2012 pattern. The yield-rate coefficients follow a pattern similar to the diagnosis rate: they are close to zero in 2010 and through 2014 and positive from 2015 onward, although they are imprecisely estimated.

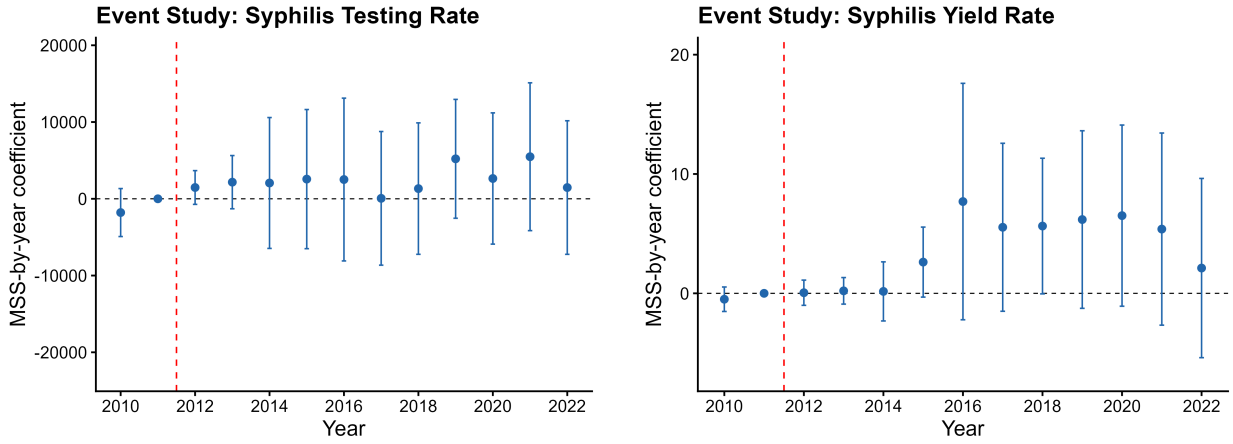


Figure 8: Event Studies: Syphilis Testing and Yield Rates

The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls. The dashed vertical line marks PrEP's 2012 FDA approval.

Table 3 reports the estimated effects on the three syphilis outcomes. Columns (1) and

(2) show a positive effect of MSS on the syphilis diagnosis rate. Without controls, it is 323.8 and significant at the 5% level. In the preferred specification with demographic controls, the coefficient is 296 and significant at the 10% level. This means a one-standard-deviation higher MSS has approximately 26.0 additional syphilis cases per 100,000 beneficiaries following PrEP's introduction. This represents an 81.5% increase relative to the 2011 baseline mean of 31.9 per 100,000. This suggests that PrEP access increased observed syphilis cases, but does not disentangle the effect of changes in testing.

The testing-rate estimates in columns (3) and (4) are positive but statistically imprecise. With controls, it is 3.986 and significant at the 10% level. The preferred estimate of 3,328.6 implies roughly 292 additional tests per 100,000 beneficiaries for a one-standard-deviation higher MSS, about 7% of the 2011 baseline mean of 4,242. This suggests that increased PrEP access did not have a large effect on syphilis testing at the population level in this sample. The yield-rate estimates in columns (5) and (6) are also positive. Without controls, the coefficient is 4.786 and significant at the 5% level. The preferred estimate implies that a one-standard-deviation higher MSS raises the yield rate by 0.35 percentage points, a 59.3% increase relative to the 2011 baseline mean of 0.59. Given that testing did not increase meaningfully, an increased testing rate implies a higher level syphilis incidence in the population.

Dependent Variables:	Diagnosis Rate		Testing Rate		Yield Rate	
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
MSS × Post	323.8**	296.3**	3,920.2	3,328.6	4.786**	3.986*
	(142.4)	(145.0)	(3,456.4)	(3,287.6)	(2.369)	(2.267)
Demographic Controls	No	Yes	No	Yes	No	Yes
2011 Baseline Mean	31.92	31.92	4,241.95	4,241.95	0.59	0.59
Fixed-effects						
State	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	647	647	647	647	647	647

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Effect of MSS × Post on Syphilis Outcomes

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic, the share of enrollees aged 0-64 who are adults aged 19-64, and the male share of enrollees aged 19-64. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

The magnitudes of the testing and yield coefficients suggest that most of the 81% increase in diagnoses is attributable to a higher share of positive tests rather than being driven by

increased syphilis detection. If the rise in diagnoses were driven entirely by more testing, testing would have had to rise by roughly the same 81% that the diagnosis rate rose. However, the 90% confidence interval for the testing rate only covers a −5% to +18% change from the 2011 baseline mean. Even at the upper end of this interval increased testing could only account for a fraction of the estimated increase in diagnoses. The remaining increase is consistent with a rise in underlying syphilis incidence. However, I cannot rule out some effect from increased testing given the imprecision of the testing and yield estimates.

5.3 Heterogeneity by Sex

Because PrEP uptake is concentrated among men, particularly men who have sex with men, the effects of PrEP access should be larger for men than in the pooled sample. I therefore estimate the main specifications separately for male and female beneficiaries. Although men disproportionately use PrEP, I do not treat women as a placebo group, because some women use PrEP themselves and changes in risky sexual behavior among men could affect women's syphilis outcomes through sexual networks. I therefore interpret differences between men and women as heterogeneity in PrEP access rather than as a placebo test.

I first examine whether the relationship between MSS and PrEP uptake differs by sex. Figure 9 shows the male and female event-study estimates side by side. For both, the 2010 coefficient is close to zero and not statistically significant relative to the omitted 2011 year. The male coefficients remain near zero through 2015 and then grow sharply. The female coefficients rise as well, but the effect is much smaller.

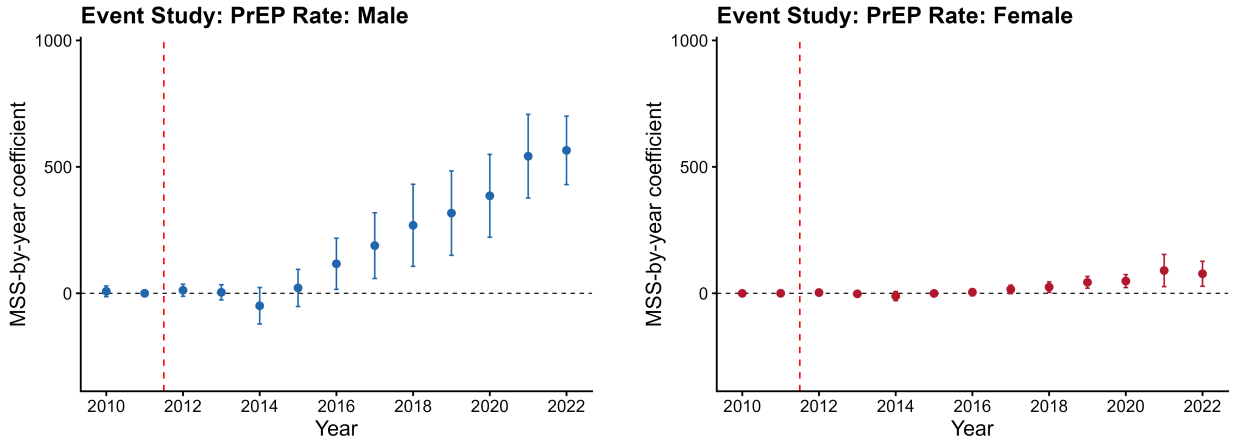


Figure 9: Event Study: PrEP Rate, Male and Female

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Male and female coefficients are estimated separately. The left panel shows male beneficiaries and the right panel shows female beneficiaries, on a common vertical axis. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3) applied to the quasi-first stage in Equation (2), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

Table 4 confirms a much stronger relationship for men. The coefficient on MSS × Post is 208.4 in the male sample and 26.35 in the female sample. The female coefficient is positive and statistically significant but roughly one-eighth the size of the male coefficient. Together, the event-study estimates and tables show that the MSS proxy captures greater PrEP uptake among men which coincides with the known proportion of PrEP use by sex.

	Male		Female	
Dependent Variable:	PrEP Rate			
Model:	(1)	(2)	(3)	(4)
Variables				
MSS × Post	245.3***	208.4***	28.86***	26.35***
	(44.01)	(41.39)	(8.060)	(9.130)
Demographic Controls	No	Yes	No	Yes
2011 Baseline Mean	6.51	6.51	2.66	2.66
Fixed-effects				
State	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	647	647	647	647

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: Quasi-First Stage: Effect of MSS × Post on the PrEP Rate, Male and Female

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Columns (1) and (2) restrict the sample to male beneficiaries and columns (3) and (4) to female beneficiaries. The male and female specifications are estimated separately. Mississippi in 2021 is excluded from the sex-specific samples because prescription claims could not be linked to beneficiary sex that year. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's male or female Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic and the share who are adults aged 19-64, computed among enrollees of the same sex as the sample. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

Figure 10 shows the male and female diagnosis-rate event-study estimates side by side. For men, the 2010 coefficient is close to zero and not statistically significant, providing no evidence of a differential change immediately prior to PrEP's introduction. After 2014, the male coefficients become positive, rising through roughly 2019 and remaining elevated thereafter. The female coefficients are smaller and less precisely estimated. For women, the 2010 coefficient is also close to zero, and the coefficients turn positive after 2014 and peak around 2019, but all post-period 95% confidence intervals include zero.

Figure 11 shows the corresponding estimates for the testing and yield rates. The male estimates are in the left column and the female estimates in the right column. The male testing-rate coefficients rise from 2018 onward. The 2019 coefficient is marginally significant at the 5% level, and the 2020-2022 coefficients are similar in size, although their 95% confidence intervals include zero. This pattern is consistent with an increase in screening for PrEP users. However, these increases are much smaller than the corresponding rise in



Figure 10: Event Study: Syphilis Diagnosis Rate, Male and Female

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Male and female coefficients are estimated separately. The left panel shows male beneficiaries and the right panel shows female beneficiaries. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

diagnoses. The female testing-rate estimates are imprecise and do not display a clear post-2012 pattern. For men, the 2010 yield coefficient is negative and its 95% confidence interval narrowly excludes zero, and the coefficients become positive after 2014. For women, the yield coefficients are close to zero through 2014 and positive afterward, though none are statistically significant.

Table 5 reports the syphilis estimates by sex. Because baseline rates differ substantially by sex, I compare effects relative to each sample's 2011 baseline mean. For the diagnosis rate, the male coefficient is positive and significant. The preferred estimate implies that a one-standard-deviation higher MSS raises the male diagnosis rate by 39.5 per 100,000, a 111.0% increase relative to the male 2011 baseline mean of 35.6. This is larger than the 81.5% increase in the pooled sample. The male testing-rate estimates are positive but not statistically significant. The coefficient corresponds to a 10.9% increase relative to the male baseline mean of 1,793 which is slightly larger in proportional terms than the pooled estimate. Its 90% confidence interval, from about -5% to $+27\%$ of the male baseline mean, likewise rules out increases in testing large enough to account for the 111.0% rise in male diagnoses. The male yield-rate coefficient is 7.896 without controls, significant at the 10% level, and 6.148 with controls, which is not statistically significant. The preferred estimate corresponds to a 36.1% increase relative to the male baseline mean of 1.57, smaller in proportional terms than the pooled estimate. For women, the syphilis estimates are positive but not statistically significant for any outcome. With controls, the female diagnosis-rate coefficient of 157.6 implies a 45.9% increase relative to the female baseline mean of 30.2, less than half

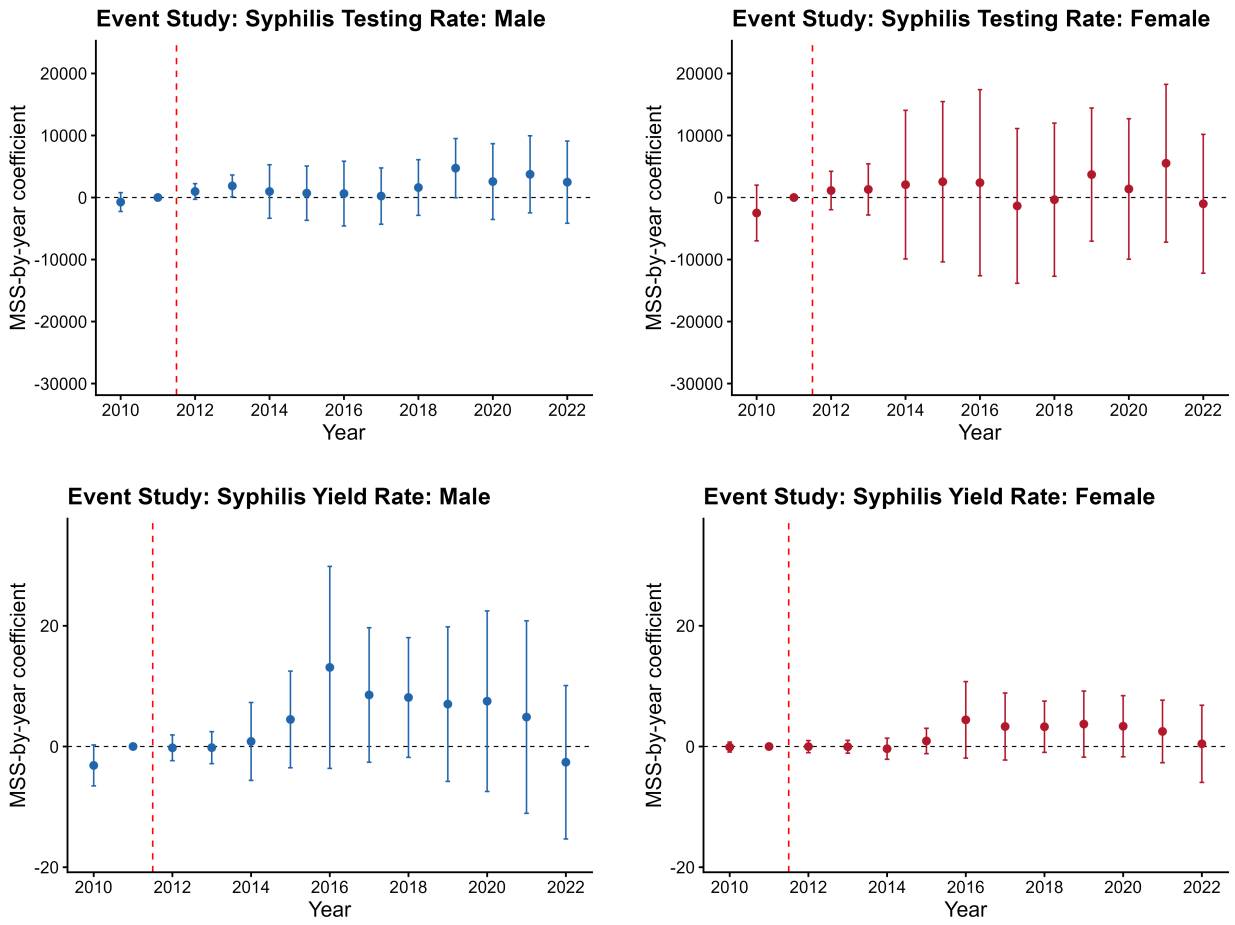


Figure 11: Event Studies: Syphilis Testing and Yield Rates, Male and Female

The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Male and female coefficients are estimated separately. The left column shows male beneficiaries and the right column shows female beneficiaries. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

of the proportional male effect, while the female yield-rate estimate corresponds to a 41.7% increase, similar in proportional terms to the male estimate.

The results by sex are consistent with the identifying logic of the MSS proxy. The relationship between MSS and PrEP uptake is substantially stronger among men, and the diagnosis effect is both larger and more precisely estimated for men than in the pooled sample. The male yield estimates are similarly imprecise, and the female estimates are positive but not statistically significant. Section 6.2 tests the difference between the male and female estimates directly.

	Male						Female					
Dependent Variables:	Diagnosis Rate		Testing Rate		Yield Rate		Diagnosis Rate		Testing Rate		Yield Rate	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables												
MSS × Post	500.5***	450.3**	2,835.6	2,223.0	7.896*	6.148	174.7	157.6	3,776.1	2,795.7	2.321	1.948
	(178.3)	(173.3)	(1,869.1)	(1,920.8)	(4.205)	(4.007)	(121.7)	(121.2)	(4,718.5)	(4,360.7)	(1.717)	(1.644)
Demographic Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
2011 Baseline Mean	35.61	35.61	1,792.95	1,792.95	1.57	1.57	30.17	30.17	5,686.71	5,686.71	0.41	0.41
Fixed-effects												
State	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics												
Observations	647	647	647	647	647	647	647	647	647	647	647	647

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 5: Effect of MSS × Post on Syphilis Outcomes, Male and Female

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. Columns (1)-(6) restrict the sample to male beneficiaries and columns (7)-(12) to female beneficiaries. The male and female specifications are estimated separately. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's male or female Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic and the share who are adults aged 19-64, computed among enrollees of the same sex as the sample. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

6 Robustness and Validity Checks

This section first assesses the identifying assumption, by bounding the effects under violations of parallel trends and by comparing men and women in a triple-difference design. It then shows that the main estimates are not driven by states with incomplete claims or by any single state.

6.1 Sensitivity to Violations of Parallel Trends

The event studies include only one pre-period coefficient, so they have limited power to detect differential pre-trends. Following Rambachan and Roth (2023), I construct confidence intervals for the average post-period effect that remain valid when parallel trends does not hold exactly. The target parameter is the average of the event-study coefficients from 2012 to 2022 in the preferred specification with demographic controls. I use the relative magnitudes restriction, which bounds the year-to-year change in the post-period violation of parallel trends by $\bar{M}$ times the change between 2010 and 2011. $\bar{M} = 0$ imposes parallel trends after 2011. $\bar{M} = 1$ allows each post-period change to be as large as the 2010-2011 change. Because the restriction applies to each year-to-year change, the allowed bias accumulates over the eleven post-period years. The breakdown value is the largest $\bar{M}$ for which the

robust confidence interval excludes zero. Table 6 and Figure 12 report the results.

The average post-period effect differs slightly from the difference-in-differences coefficient because the latter compares the post-period years with both 2010 and 2011. For the PrEP rate, the robust confidence intervals for the pooled and male samples exclude zero for every value of $\bar{M}$ up to 1. The quasi-first-stage results therefore hold even if the post-period violation of parallel trends changes each year by as much as the 2010-2011 change. The female PrEP estimate has a breakdown value of 0.4. For the diagnosis rate, the conventional 95% confidence interval for the average post-period effect excludes zero in the pooled and male samples, and the breakdown value is 0.2 in both. Under $\bar{M} = 0.2$, the violation of parallel trends can grow by 0.2 times the 2010-2011 change in each post-period year, or 2.2 times that change by 2022. For the testing and yield rates, and for the female diagnosis rate, the conventional 95% confidence interval for the average post-period effect already includes zero, so there is no breakdown value to report. This includes the pooled yield rate, whose difference-in-differences estimate is significant at the 10% level but whose average post-period effect has a 95% confidence interval of −0.47 to 8.13.

		Average		Robust 95% CI		Breakdown
Outcome	Sample	2012-2022 effect	95% CI	$\bar{M} = 0.5$	$\bar{M} = 1$	$\bar{M}$
PrEP	Pooled	114.2	[73.2, 155.2]	[51.0, 180.8]	[19.2, 215.9]	≥ 1.0
	Male	215.6	[131.7, 299.5]	[96.4, 343.3]	[36.5, 406.7]	≥ 1.0
	Female	26.6	[10.4, 42.8]	[-1.1, 56.2]	[-16.3, 73.4]	0.4
Diagnosis	Pooled	287.5	[18.5, 556.5]	[-56.0, 790.4]	[-286.8, 1,070.7]	0.2
	Male	417.7	[84.8, 750.7]	[-191.0, 1,162.5]	[-673.8, 1,665.7]	0.2
	Female	151.1	[-86.2, 388.3]	[-195.4, 512.1]	[-379.6, 701.1]	–
Testing	Pooled	2,450.9	[-4,077.1, 8,978.9]	[-12,418.7, 17,854.0]	[-24,687.8, 30,123.1]	–
	Male	1,864.4	[-2,019.3, 5,748.1]	[-5,474.6, 9,044.6]	[-10,949.0, 14,439.8]	–
	Female	1,574.3	[-7,570.1, 10,718.8]	[-20,936.4, 21,656.5]	[-38,496.7, 39,030.0]	–
Yield	Pooled	3.83	[-0.47, 8.13]	[-1.31, 11.16]	[-3.95, 14.85]	–
	Male	4.68	[-3.32, 12.69]	[-16.17, 25.04]	[-32.84, 41.55]	–
	Female	1.96	[-1.12, 5.03]	[-2.22, 6.51]	[-3.79, 8.64]	–

Table 6: Robust Confidence Intervals for the Average Post-Period Effect

The average 2012-2022 effect is the mean of the event-study coefficients from Equation (3) for 2012-2022, from the preferred specification with demographic controls, with 2011 as the omitted reference year. The 95% CI is the conventional interval for that average. Robust 95% CIs are from Rambachan and Roth (2023) under the relative magnitudes restriction, which bounds each post-period year-to-year change in the violation of parallel trends by $\bar{M}$ times the 2010-2011 change. Intervals reaching the search grid of ±20 standard errors are reported as unbounded. The breakdown $\bar{M}$ is the largest value on a grid from 0 to 1 in steps of 0.1 for which the robust CI excludes zero. "–" indicates that the robust CI includes zero even at $\bar{M} = 0$. Male and female coefficients are estimated separately.

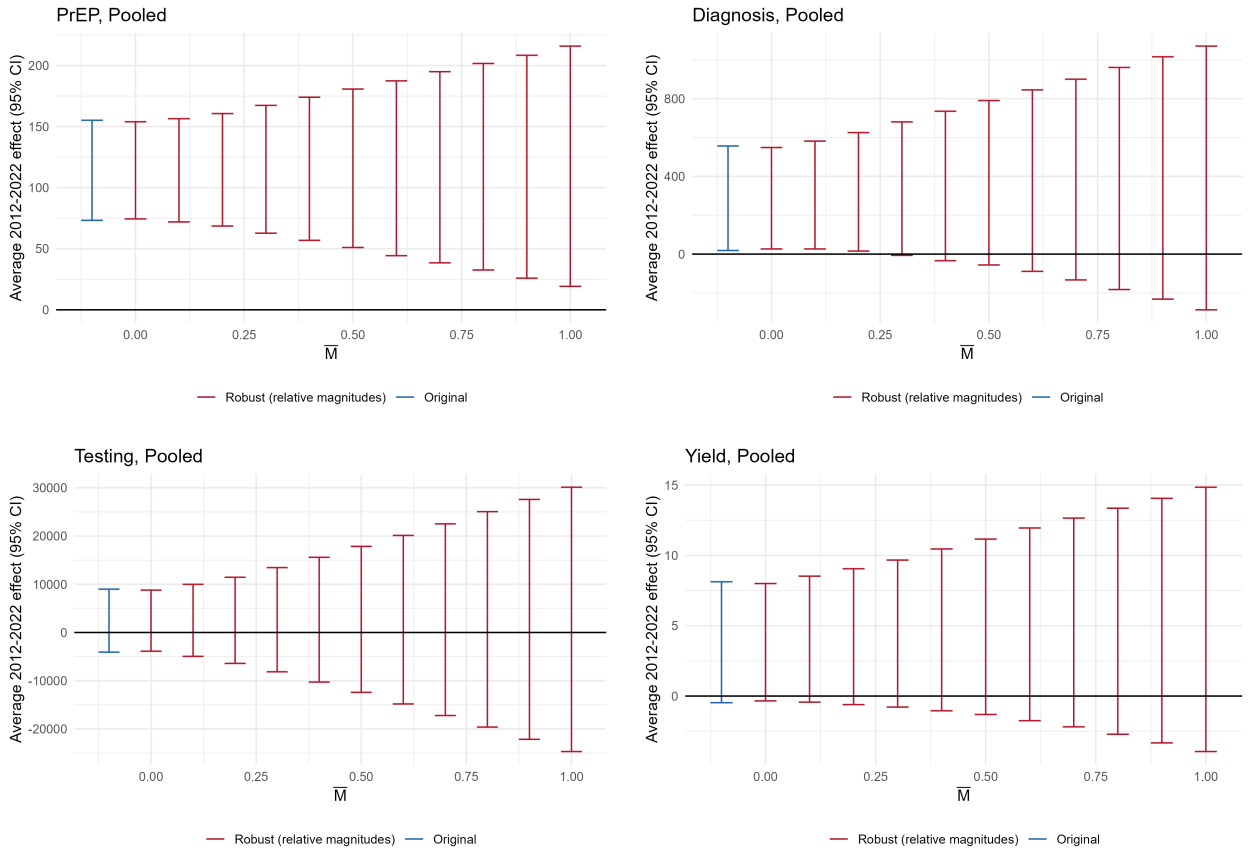


Figure 12: Sensitivity of the Average Post-Period Effect to Violations of Parallel Trends

Each panel shows the conventional 95% CI for the average 2012-2022 event-study coefficient (Original) and the Rambachan and Roth (2023) robust 95% CI under the relative magnitudes restriction for values of $\bar{M}$ from 0 to 1. Panels show the PrEP, syphilis diagnosis, testing, and yield rates in the pooled sample. Event-study estimates shown are from the specification with demographic controls.

6.2 Triple-Difference Estimates

This section reports the triple-difference estimates from Equation (4), which test whether the effects of MSS differ between men and women.

Quasi-first stage. Figure 13 shows the triple-difference event-study estimates for the PrEP rate. Each coefficient is the difference between the male and female MSS-by-year effects, relative to 2011. Table 7 reports $\hat{\delta}$.

The 2010 coefficient is close to zero, and the coefficients remain near zero through 2014. From 2015 onward they rise steadily, reaching roughly 500 by 2022, and their 95% confidence intervals exclude zero. In Table 7, $\hat{\delta}$ is 216.4 without controls and 211.9 with controls, both significant at the 1% level. With controls, a one-standard-deviation higher MSS raises the male PrEP rate by about 18.6 more patients per 100,000 than the female rate. The female effect in the same model is 13.78 with controls and is not statistically significant, compared with 26.35 when the female sample is estimated separately in Table 4. The triple difference

therefore confirms that the MSS proxy predicts PrEP uptake primarily among men.

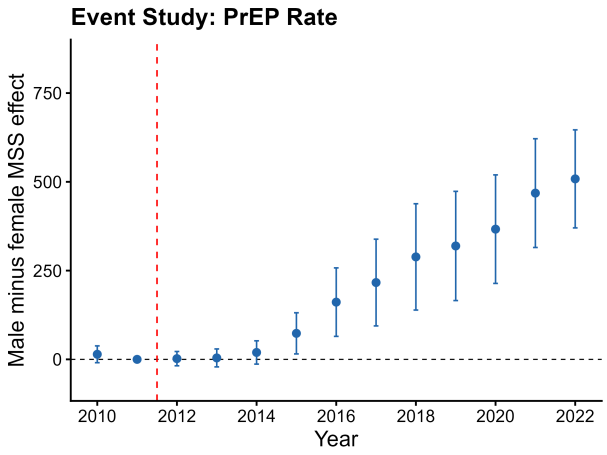


Figure 13: Triple-Difference Event Study: PrEP Rate

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Coefficients and 95% confidence intervals are the differences between the male and female MSS-by-year coefficients from the stacked male and female panels, with state-by-sex and sex-by-year fixed effects and 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

Dependent Variable:	PrEP Rate	
Model:	(1)	(2)
Variables		
MSS × Post	28.86***	13.78
	(8.063)	(13.25)
MSS × Post × Male	216.4***	211.9***
	(43.65)	(43.29)
Demographic Controls	No	Yes
2011 Baseline Mean	4.59	4.59
Fixed-effects		
State-Sex	Yes	Yes
Year-Sex	Yes	Yes
Fit statistics		
Observations	1,294	1,294

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 7: Triple-Difference Estimates: Effect of MSS × Post × Male on the PrEP Rate

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The MSS × Post × Male row reports $\hat{\delta}$ from Equation (4), and the MSS × Post row reports the female effect. The models are estimated on stacked male and female state-year panels with state-by-sex and sex-by-year fixed effects. Standard errors clustered at the state level are reported in parentheses. Specifications marked "Yes" in the Demographic Controls row include the sex-specific demographic controls and the Medicaid expansion indicator described in Section 3.

Syphilis. Figures 14 and 15 show the triple-difference event-study estimates for the diagnosis, testing, and yield rates. Table 8 reports $\hat{\delta}$ for each outcome.

For the diagnosis rate, the 2010 coefficient is negative but its 95% confidence interval includes zero, and the coefficients are close to zero in 2012 and 2013. From 2014 onward they

are positive, rise through 2021, and their 95% confidence intervals exclude zero. The testing-rate coefficients are imprecise and show no post-2012 pattern. The yield-rate coefficients are positive from 2014 through 2021, although most individual 95% confidence intervals include zero, and the 2010 coefficient is negative with an upper confidence bound near zero.

In Table 8, the male differential in the diagnosis rate is 325.8 without controls and 321.4 with controls, both significant at the 1% level. With controls, a one-standard-deviation higher MSS raises the male diagnosis rate by about 28.2 more cases per 100,000 than the female rate, 79.2% of the male 2011 baseline mean of 35.6. This is similar to the difference of 292.7 between the separately estimated male and female coefficients in Table 5. The male differential in the yield rate is 5.575 without controls and 5.490 with controls, both significant at the 5% level. The preferred estimate corresponds to a 0.48 percentage-point larger increase in the male yield rate, 30.7% of the male baseline mean of 1.57. The yield differential is more precisely estimated than the separate male yield estimate, which is not statistically significant with controls. The testing-rate differential is negative and not statistically significant, −940.5 without controls and −1,265.1 with controls. Testing therefore did not rise more among men than among women in high-MSS states, while diagnoses and the share of positive tests did. Because women also use PrEP, these differentials understate the full effect of PrEP access on men.

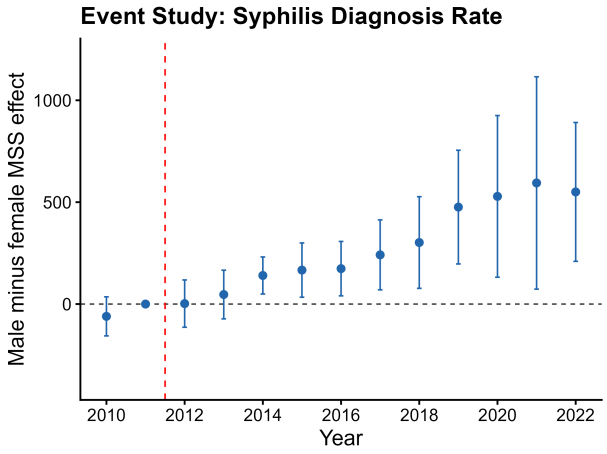


Figure 14: Triple-Difference Event Study: Syphilis Diagnosis Rate

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Coefficients and 95% confidence intervals are the differences between the male and female MSS-by-year coefficients from the stacked male and female panels, with state-by-sex and sex-by-year fixed effects and 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

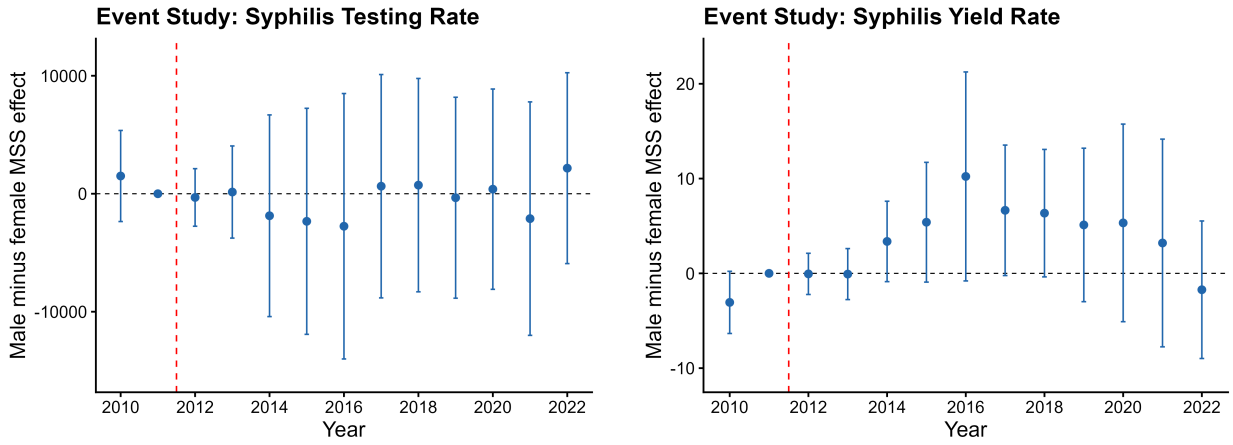


Figure 15: Triple-Difference Event Studies: Syphilis Testing and Yield Rates

The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. The left panel shows the testing rate and the right panel shows the yield rate. Coefficients and 95% confidence intervals are the differences between the male and female MSS-by-year coefficients from the stacked male and female panels, with state-by-sex and sex-by-year fixed effects and 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

Dependent Variables:	Diagnosis Rate		Testing Rate		Yield Rate	
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
MSS × Post	174.7	145.1	3,776.1	3,038.7	2.321	1.333
	(121.8)	(119.1)	(4,720.3)	(4,479.4)	(1.718)	(1.680)
MSS × Post × Male	325.8***	321.4***	-940.5	-1,265.1	5.575**	5.490**
	(70.82)	(73.79)	(3,376.5)	(3,405.3)	(2.677)	(2.617)
Demographic Controls	No	Yes	No	Yes	No	Yes
2011 Baseline Mean	32.89	32.89	3,739.83	3,739.83	0.99	0.99
Fixed-effects						
State-Sex	Yes	Yes	Yes	Yes	Yes	Yes
Year-Sex	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	1,294	1,294	1,294	1,294	1,294	1,294

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8: Triple-Difference Estimates: Effect of MSS × Post × Male on Syphilis Outcomes

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. The MSS × Post × Male row reports $\hat{\delta}$ from Equation (4), and the MSS × Post row reports the female effect. The models are estimated on stacked male and female state-year panels with state-by-sex and sex-by-year fixed effects. Standard errors clustered at the state level are reported in parentheses. Specifications marked "Yes" in the Demographic Controls row include the sex-specific demographic controls and the Medicaid expansion indicator described in Section 3.

6.3 Excluding States with Incomplete Claims

Kansas and Maine are absent from the MAX files in 2010 because of data-reporting issues, Idaho appears in the prescription files in 2011 but not in the outpatient files, and Mississippi's 2021 claims cannot be linked to beneficiaries. These states are included in the main analysis, but I re-estimate the specifications after excluding all four to verify that the results are not driven by these data limitations.

Figure 16 shows that the event-study estimates for the PrEP rate in this restricted sample follow the same pattern as in the full sample.

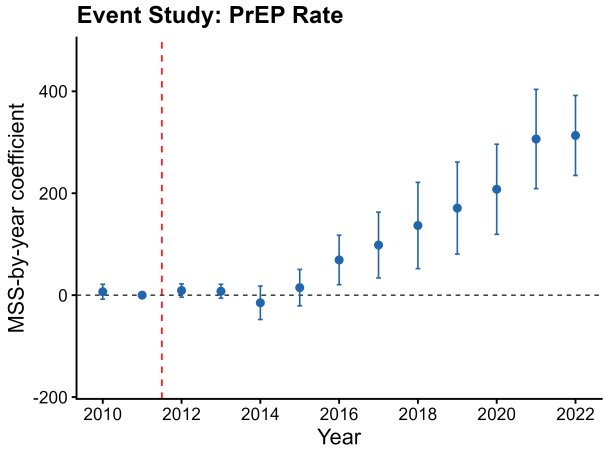


Figure 16: Event Study: PrEP Rate, Excluding Kansas, Maine, Idaho, and Mississippi

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Sample excludes Kansas, Maine, Idaho, and Mississippi. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3) applied to the quasi-first stage in Equation (2), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

Table 9 shows a similarly positive and statistically significant relationship between MSS and PrEP uptake. With demographic controls, the coefficient on MSS $\times$ Post is 116.1 ($p < 0.01$), compared with 112.1 in the full sample.

Figure 17 presents the event-study estimates for the diagnosis rate in the restricted sample. As in the full sample, the 2010 coefficient is close to zero and not statistically significant relative to the omitted 2011 year, and the coefficients become positive after 2014.

Figure 18 presents the corresponding estimates for the testing and yield rates. The testing coefficients remain imprecisely estimated and do not display a clear post-2012 pattern, while the yield coefficients follow the same pattern as in the full sample.

Table 10 shows that the syphilis estimates are very similar to the full-sample results. With demographic controls, the diagnosis-rate coefficient is 298.9, compared with 296.3 in the full sample, and is statistically significant at the 10% level, compared with the 5% level in the full sample. The testing-rate coefficient is 2,309.8, compared with 3,328.6, and remains

Dependent Variable:	PrEP Rate	
Model:	(1)	(2)
Variables		
MSS × Post	130.1***	116.1***
	(22.13)	(20.12)
Demographic Controls	No	Yes
2011 Baseline Mean	4.14	4.14
Fixed-effects		
State	Yes	Yes
Year	Yes	Yes
Fit statistics		
Observations	598	598

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 9: Quasi-First Stage: Effect of MSS × Post on the PrEP Rate, Excluding Kansas, Maine, Idaho, and Mississippi

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Sample excludes Kansas, Maine, Idaho, and Mississippi. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic, the share of enrollees aged 0-64 who are adults aged 19-64, and the male share of enrollees aged 19-64. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

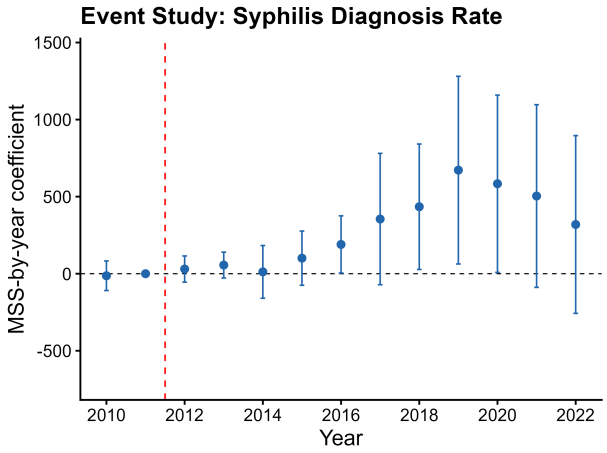


Figure 17: Event Study: Syphilis Diagnosis Rate, Excluding Kansas, Maine, Idaho, and Mississippi

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Sample excludes Kansas, Maine, Idaho, and Mississippi. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

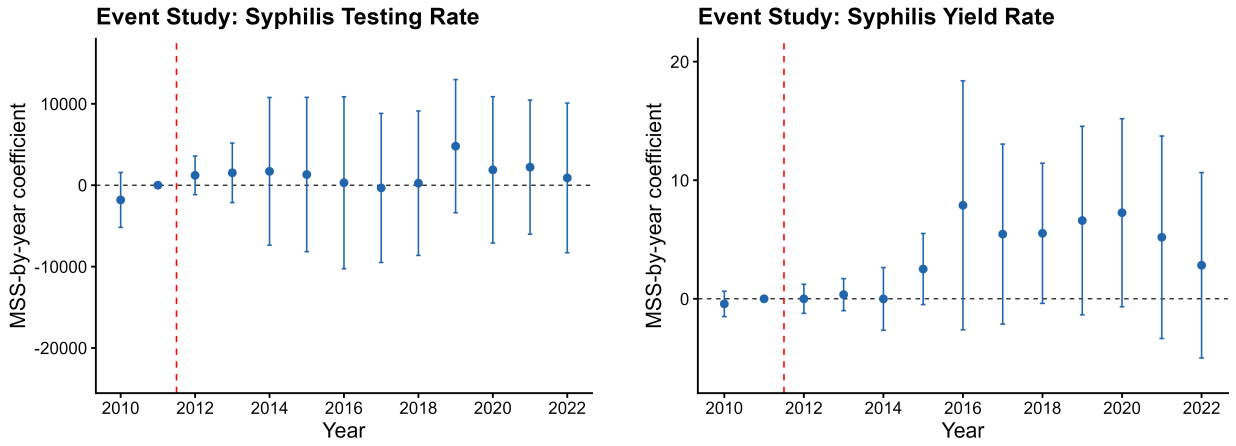


Figure 18: Event Studies: Syphilis Testing and Yield Rates, Excluding Kansas, Maine, Idaho, and Mississippi

The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Sample excludes Kansas, Maine, Idaho, and Mississippi. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

statistically imprecise in both specifications. The yield-rate coefficient is 4.134, compared with 3.986, and remains significant at the 10% level. Without controls it is 4.775, also significant at the 10% level. The close similarity of the estimates and event-study patterns across samples indicates that the main syphilis results are not driven by the incomplete claims observations for Kansas, Maine, Idaho, or Mississippi.

Tables 11 and 12 and Figure 19 repeat the sex-specific analysis in Section 5.3 for this restricted sample. The quasi-first-stage estimates are similar to those in the full sample. With controls, the male coefficient is 214.5, compared with 208.4, and the female coefficient is 27.38, compared with 26.35. All four estimates remain significant at the 1% level. The male diagnosis-rate coefficient with controls is 451.5, compared with 450.3, and remains significant at the 5% level. It implies a 107.9% increase relative to the male 2011 baseline mean of 36.75 for a one-standard-deviation higher MSS. The male testing-rate coefficient is 1,819.4, an 8.8% increase relative to the male baseline mean, and is not statistically significant. The male yield-rate coefficient is 7.708 without controls, significant at the 10% level, and 6.062 with controls, which is not statistically significant. The female estimates remain positive and not statistically significant for every outcome. With controls, the female diagnosis-rate coefficient of 155.4 implies a 45.4% increase relative to the female baseline mean of 30.08. The event-study estimates in Figure 19 follow the same patterns as in the full sample.

Dependent Variables:	Diagnosis Rate		Testing Rate		Yield Rate	
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
MSS × Post	321.2**	298.9*	2,814.6	2,309.8	4.775*	4.134*
	(148.3)	(154.0)	(3,523.0)	(3,452.2)	(2.466)	(2.427)
Demographic Controls	No	Yes	No	Yes	No	Yes
2011 Baseline Mean	32.29	32.29	4,218.34	4,218.34	0.61	0.61
Fixed-effects						
State	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	598	598	598	598	598	598

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 10: Effect of MSS × Post on Syphilis Outcomes, Excluding Kansas, Maine, Idaho, and Mississippi

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. Sample excludes Kansas, Maine, Idaho, and Mississippi. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic, the share of enrollees aged 0-64 who are adults aged 19-64, and the male share of enrollees aged 19-64. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

	Male		Female	
Dependent Variable:	PrEP Rate			
Model:	(1)	(2)	(3)	(4)
Variables				
MSS × Post	246.4***	214.5***	29.61***	27.38***
	(46.61)	(43.73)	(8.479)	(9.607)
Demographic Controls	No	Yes	No	Yes
2011 Baseline Mean	6.51	6.51	2.90	2.90
Fixed-effects				
State	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	598	598	598	598

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 11: Quasi-First Stage: Effect of MSS × Post on the PrEP Rate, Male and Female, Excluding Kansas, Maine, Idaho, and Mississippi

The PrEP rate is the number of patients with a PrEP prescription per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Sample excludes Kansas, Maine, Idaho, and Mississippi. Columns (1) and (2) restrict the sample to male beneficiaries and columns (3) and (4) to female beneficiaries. The male and female specifications are estimated separately. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's male or female Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic and the share who are adults aged 19-64, computed among enrollees of the same sex as the sample. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

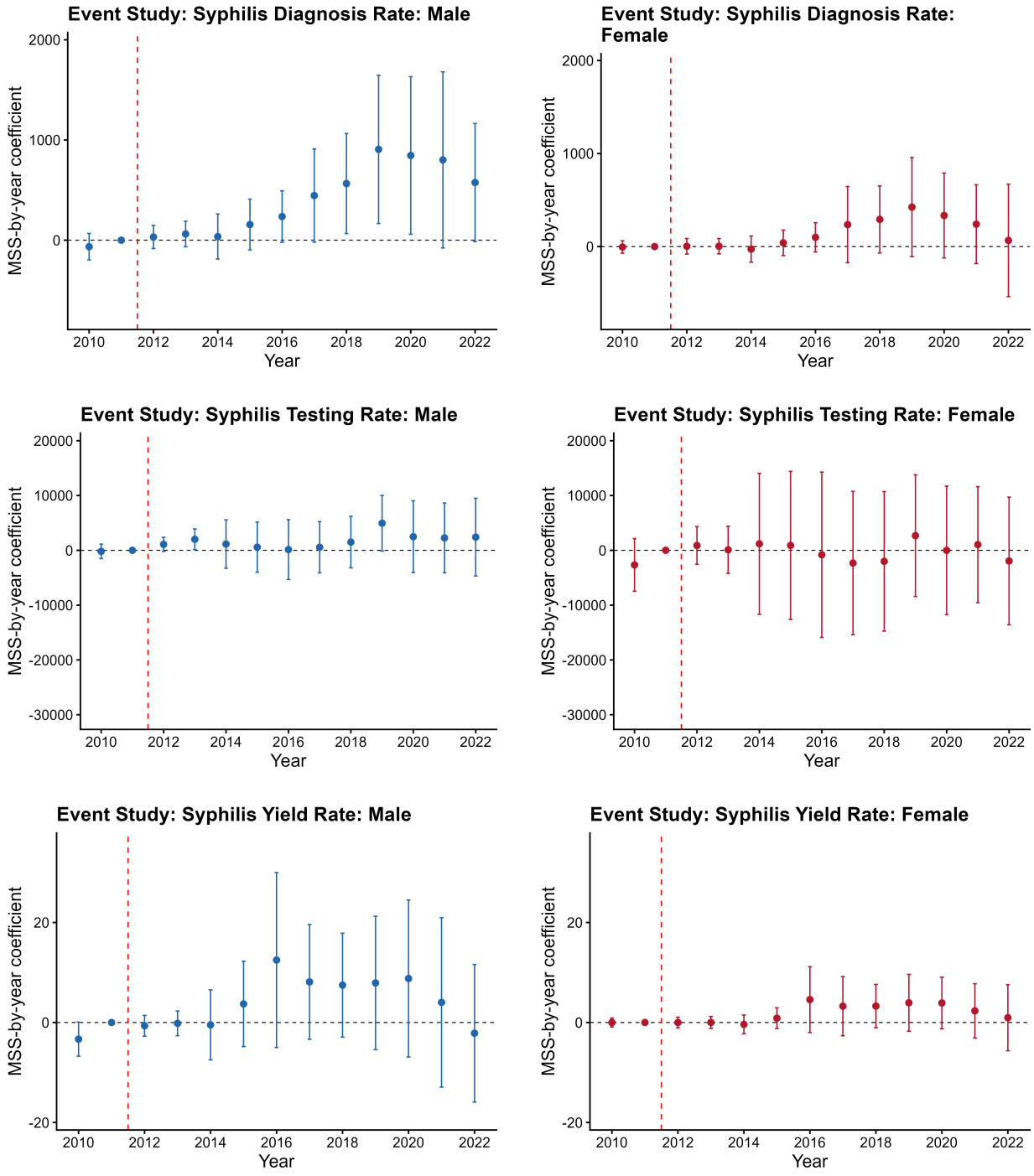


Figure 19: Event Studies: Syphilis Outcomes, Male and Female, Excluding Kansas, Maine, Idaho, and Mississippi

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Sample excludes Kansas, Maine, Idaho, and Mississippi. Male and female coefficients are estimated separately. The left column shows male beneficiaries and the right column shows female beneficiaries. Rows show the diagnosis, testing, and yield rates. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

	Male						Female					
Dependent Variables:	Diagnosis Rate		Testing Rate		Yield Rate		Diagnosis Rate		Testing Rate		Yield Rate	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Variables												
MSS × Post	490.3**	451.5**	2,232.6	1,819.4	7.708*	6.062	175.9	155.4	2,258.9	1,273.3	2.374	2.043
	(185.9)	(183.4)	(1,947.8)	(2,051.2)	(4.415)	(4.241)	(126.9)	(126.0)	(4,827.6)	(4,500.4)	(1.787)	(1.728)
Demographic Controls	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
2011 Baseline Mean	36.75	36.75	1,811.67	1,811.67	1.64	1.64	30.08	30.08	5,638.23	5,638.23	0.42	0.42
Fixed-effects												
State	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics												
Observations	598	598	598	598	598	598	598	598	598	598	598	598

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 12: Effect of MSS × Post on Syphilis Outcomes, Male and Female, Excluding Kansas, Maine, Idaho, and Mississippi

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. Sample excludes Kansas, Maine, Idaho, and Mississippi. Columns (1)-(6) restrict the sample to male beneficiaries and columns (7)-(12) to female beneficiaries. The male and female specifications are estimated separately. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's male or female Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic and the share who are adults aged 19-64, computed among enrollees of the same sex as the sample. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

6.4 Leave-One-Out

The identifying variation in the main specification comes from cross-state differences in the pre-treatment MSS share, so it is also important to verify that the estimated effects are not driven by a single influential state. I therefore re-estimate the preferred specification 50 times, excluding one state from the sample in each iteration. Figures 20 and 21 report the resulting coefficient estimates with 90% and 95% confidence intervals for the diagnosis, testing, and yield rates.

The diagnosis and yield rate point estimates remain positive in every specification, and for most exclusions they lie close to the full-sample estimates of 296.3 and 3.986. The largest declines occur when Nevada or Delaware is omitted. Omitting Nevada roughly halves the diagnosis-rate estimate, and omitting either state roughly halves the yield-rate estimate. Both states have among the highest MSS shares in the sample and therefore carry substantial leverage. Omitting California produces the largest estimates for both outcomes. Because the full-sample yield estimate is significant at the 10% level but not the 5% level, the figures show both 90% and 95% intervals. The 90% intervals correspond to the significance stars in the tables. The 90% interval for the diagnosis rate excludes zero for every exclusion except

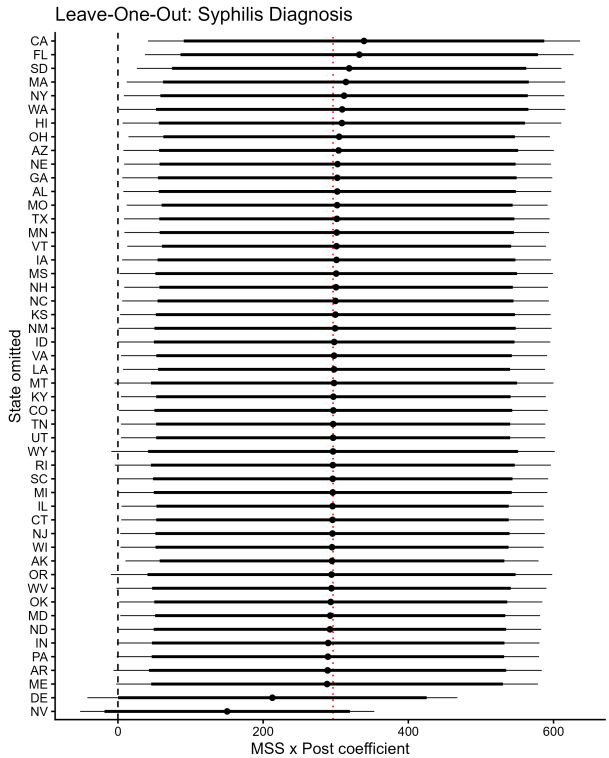


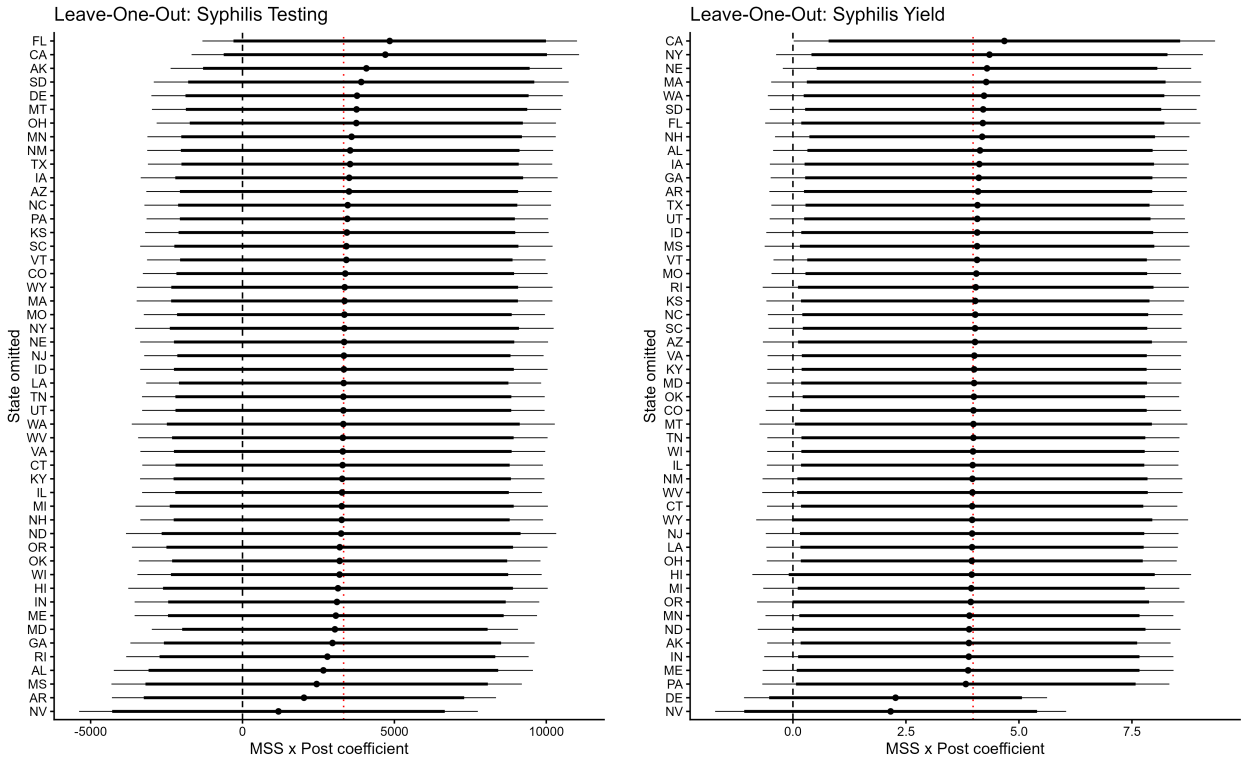
Figure 20: Leave-One-Out Estimates: Syphilis Diagnosis Rate

The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Each row reports the MSS $\times$ Post coefficient from the preferred specification with demographic controls, re-estimated after excluding the labeled state. Thick bars show 90% confidence intervals and thin lines show 95% confidence intervals, using critical values from a t distribution with $G - 1$ degrees of freedom. The dotted vertical line marks the full-sample estimate.

Delaware and Nevada. The full-sample yield estimate with controls is significant only at the 10% level, so the 90% yield intervals lie close to zero for most exclusions and include zero when Delaware or Nevada is omitted.

The testing-rate point estimates are also positive across all state exclusions. They range from their smallest value when Nevada is omitted to their largest when Florida is omitted. Every 95% confidence interval includes zero. This suggests that the imprecision of the testing estimates is not attributable to any single state.

Figures 22, 23, and 24 repeat the leave-one-out exercise separately for male and female beneficiaries, using the preferred specification with demographic controls. All 50 male diagnosis-rate estimates are positive. Excluding any single state other than Nevada or Delaware leaves the estimate between roughly 415 and 490 which is close to the full-sample estimate of 450.3. As in the pooled sample, omitting Nevada or Delaware produces the largest declines, to roughly 265 and 325. The 90% interval excludes zero for every exclusion, and the 95% interval reaches zero only when Nevada is omitted. The male testing and yield rate point estimates are also positive for every exclusion. The yield estimates are again



(a) Syphilis Testing Rate

(b) Syphilis Yield Rate

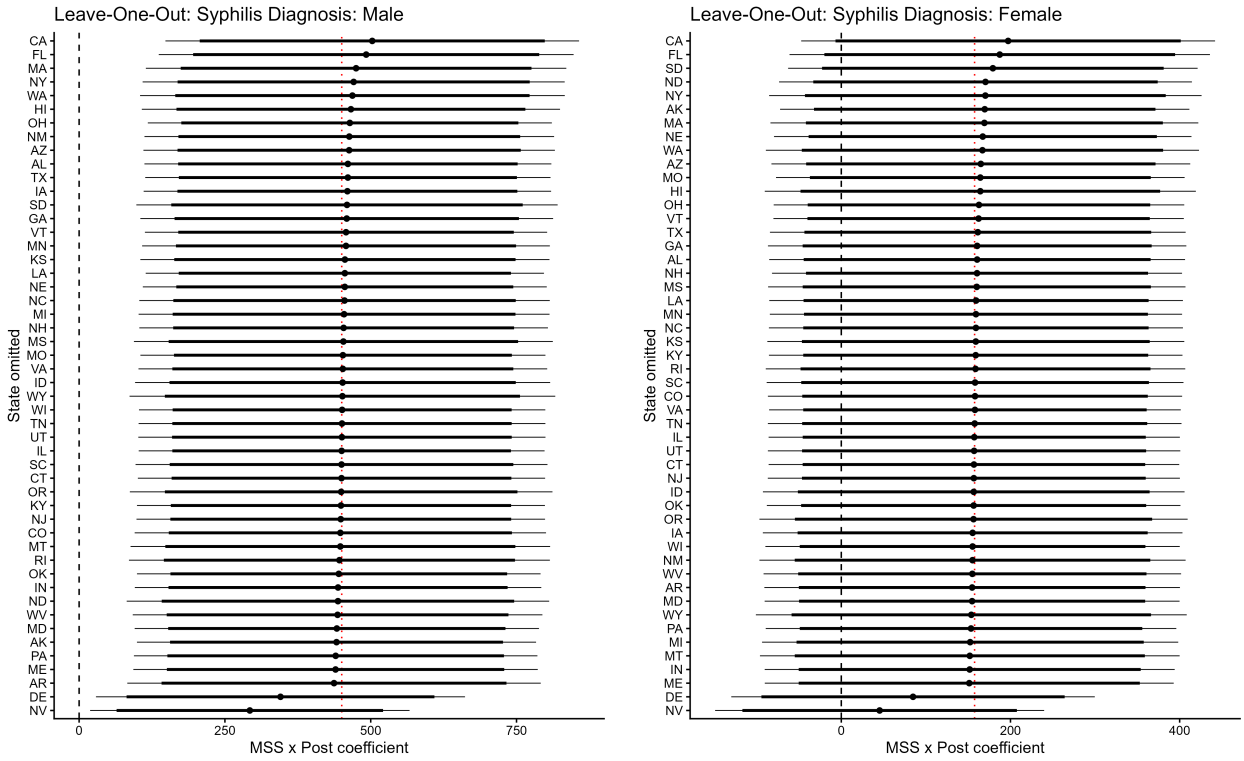
Figure 21: Leave-One-Out Estimates: Testing and Yield Rates

The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Each row reports the MSS $\times$ Post coefficient from the preferred specification with demographic controls, re-estimated after excluding the labeled state. Thick bars show 90% confidence intervals and thin lines show 95% confidence intervals, using critical values from a t distribution with $G-1$ degrees of freedom. The dotted vertical line marks the full-sample estimate.

smallest when Delaware or Nevada is omitted. Additionally, the testing and yield intervals generally include zero, as in the full male sample.

All 50 female diagnosis-rate estimates are positive, and every 90% interval includes zero, as in the full female sample. As in the pooled and male samples, omitting Nevada or Delaware produces the largest declines, to roughly 35 and 75, and omitting California produces the largest estimate. The female yield estimates are also positive for every exclusion and are smallest when Delaware or Nevada is omitted. The female testing and yield intervals include zero, as in the full female sample.

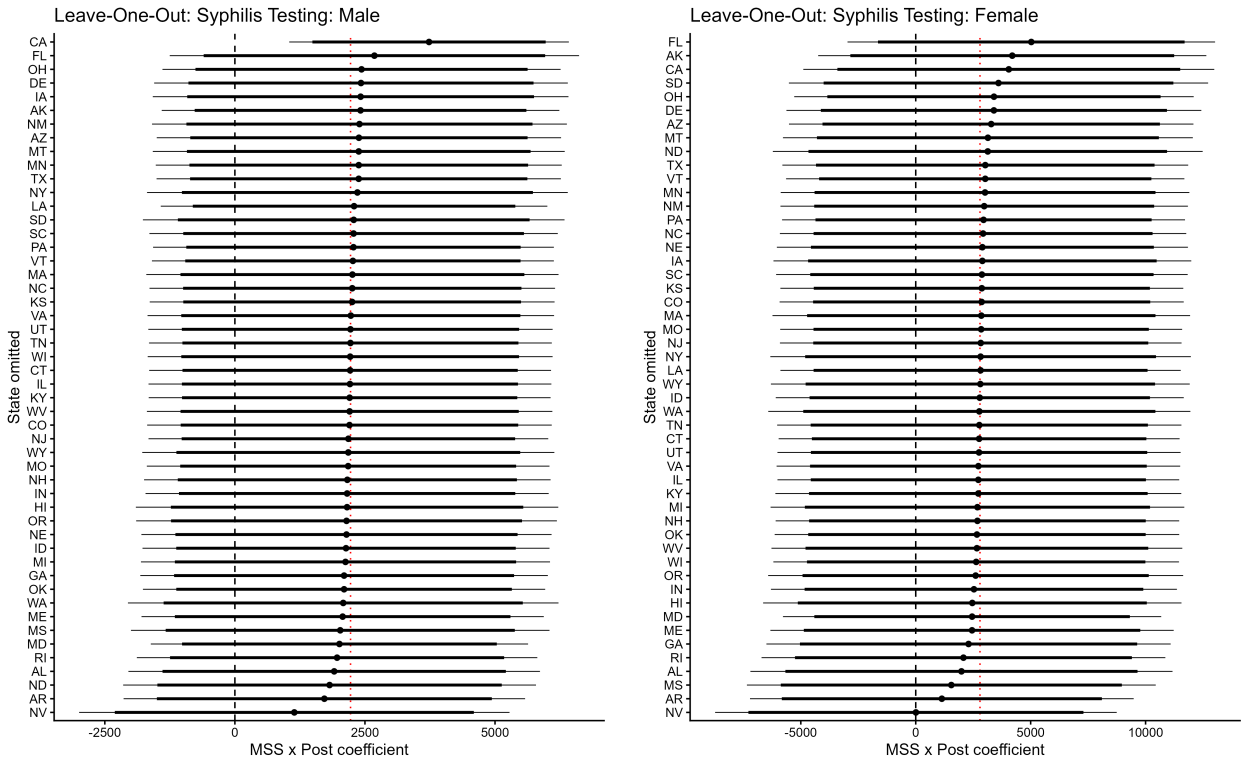
Overall, the leave-one-out exercise shows that the sign of the diagnosis and yield effects does not depend on any single state, in the pooled sample or for either sex, while also showing that their magnitude is sensitive to the inclusion of Delaware and Nevada.



(a) Male (b) Female

Figure 22: Leave-One-Out Estimates by Sex: Syphilis Diagnosis Rate

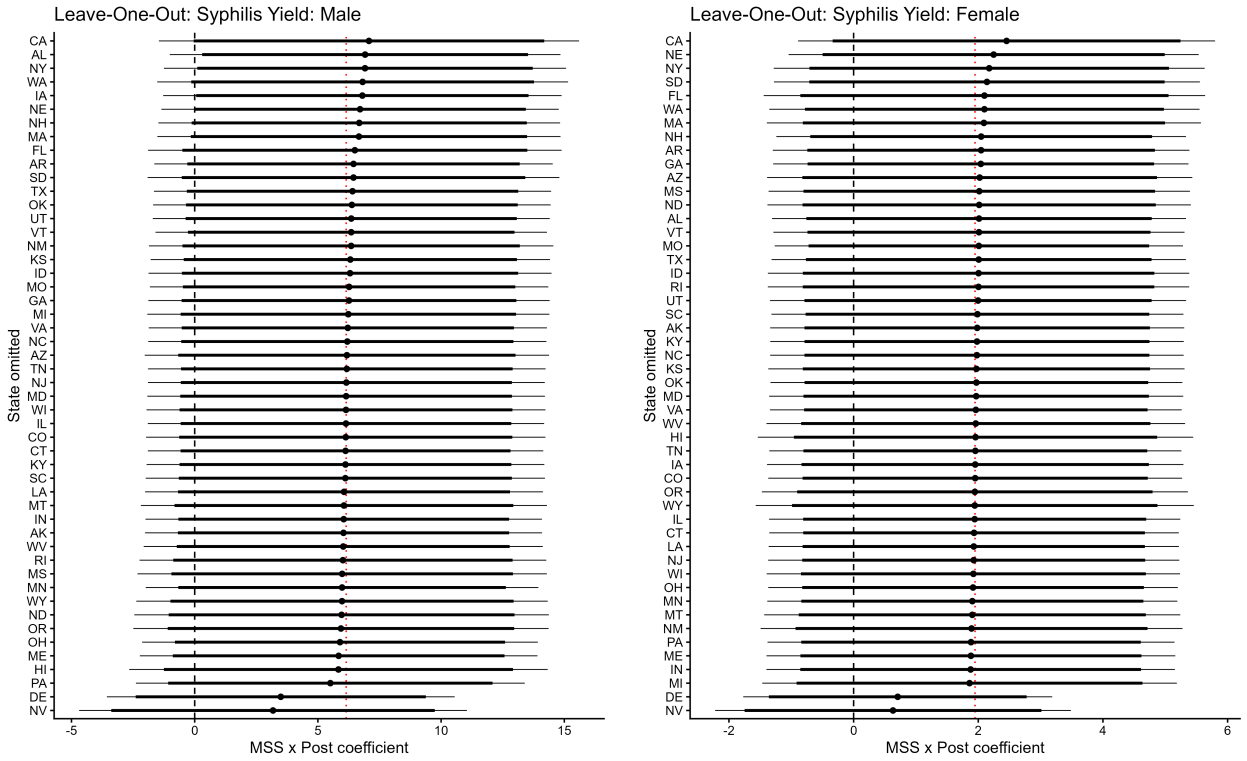
The diagnosis rate is the number of syphilis cases per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Male and female coefficients are estimated separately. Each row reports the MSS $\times$ Post coefficient from the preferred specification with demographic controls, re-estimated after excluding the labeled state. Thick bars show 90% confidence intervals and thin lines show 95% confidence intervals, using critical values from a t distribution with $G - 1$ degrees of freedom. The dotted vertical line marks the full-sample estimate.



(a) Male (b) Female

Figure 23: Leave-One-Out Estimates by Sex: Syphilis Testing Rate

The testing rate is the number of syphilis test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Male and female coefficients are estimated separately. Each row reports the MSS × Post coefficient from the preferred specification with demographic controls, re-estimated after excluding the labeled state. Thick bars show 90% confidence intervals and thin lines show 95% confidence intervals, using critical values from a t distribution with $G - 1$ degrees of freedom. The dotted vertical line marks the full-sample estimate.



(a) Male (b) Female

Figure 24: Leave-One-Out Estimates by Sex: Syphilis Yield Rate

The yield rate is the number of syphilis cases divided by the total number of syphilis test events in the state, multiplied by 100, measured annually. A case is a test followed within 30 days by a syphilis diagnosis, counted at most once per beneficiary in any 90-day period. Male and female coefficients are estimated separately. Each row reports the MSS × Post coefficient from the preferred specification with demographic controls, re-estimated after excluding the labeled state. Thick bars show 90% confidence intervals and thin lines show 95% confidence intervals, using critical values from a t distribution with $G - 1$ degrees of freedom. The dotted vertical line marks the full-sample estimate.

7 Conclusion

PrEP has transformed HIV prevention by substantially reducing the risk of acquiring HIV, but this may lead to risk compensation where individuals who are less afraid of getting HIV are more likely to engage in risky sexual behavior. Because PrEP does not prevent bacterial STIs, an increase in risky behavior induced by PrEP may lead to increased syphilis spread. At the same time, PrEP users are recommended to undergo more frequent screening for syphilis meaning there could have been a coinciding increase in the detection of previously undiagnosed syphilis. In this paper, I address that distinction by using Medicaid claims to observe both syphilis diagnoses and syphilis testing.

I exploit cross-state variation in the pre-treatment share of cohabiting male same-sex couple households as a proxy for differential PrEP access following its 2012 FDA approval. The quasi-first-stage results show that states with higher MSS experienced greater subsequent PrEP uptake, with a substantially stronger relationship among men. I then estimate effects on the syphilis diagnosis rate, testing rate, and yield rate using a difference-in-differences framework with state and year fixed effects.

The results show increases in syphilis diagnoses in states with greater PrEP access. Greater PrEP access, proxied by a one-standard-deviation higher MSS, raises the syphilis diagnosis rate by 81.5% relative to its 2011 baseline mean, an estimate significant at the 5% level in the preferred specification. The estimated effect on testing is imprecise and too small to account for the rise in diagnoses. The point estimate for the yield rate is a 59.3% increase, which in conjunction with little to no increase in testing, suggests that more syphilis tests are coming back positive. Together, these three estimates suggest that most of the rise in diagnosed syphilis reflects a higher share of positive tests rather than more testing. This is consistent with risk compensation following PrEP's introduction inducing an increase in underlying syphilis incidence. At the same time, the imprecision of the yield estimate means that the analysis cannot determine precisely how much increased screening contributed to the observed increase in diagnoses.

Several additional results support this interpretation. The event-study estimates show no differential change between 2010 and 2011, and show diagnosis and yield effects emerging after 2014, in step with the delayed rise in PrEP uptake. The relationship between MSS and both PrEP uptake and syphilis diagnoses is substantially stronger among men, consistent with the concentration of PrEP use among men. The estimates are nearly unchanged after excluding states with incomplete claims observations, and leave-one-out specifications show that the positive diagnosis and yield point estimates do not depend on any single state, although their magnitude is sensitive to the inclusion of Delaware and Nevada.

The analysis is limited to Medicaid beneficiaries, so the magnitude of these effects may differ in populations covered by private insurance or other sources of care. In addition, MSS is a proxy for differential PrEP access rather than a direct measure of PrEP use, although the relationship between MSS and observed PrEP uptake supports its use in the empirical design. Future work using broader claims data could test whether similar patterns appear in other insured populations and further characterize how changes in preventive technologies affect both health behavior and the detection of related conditions.

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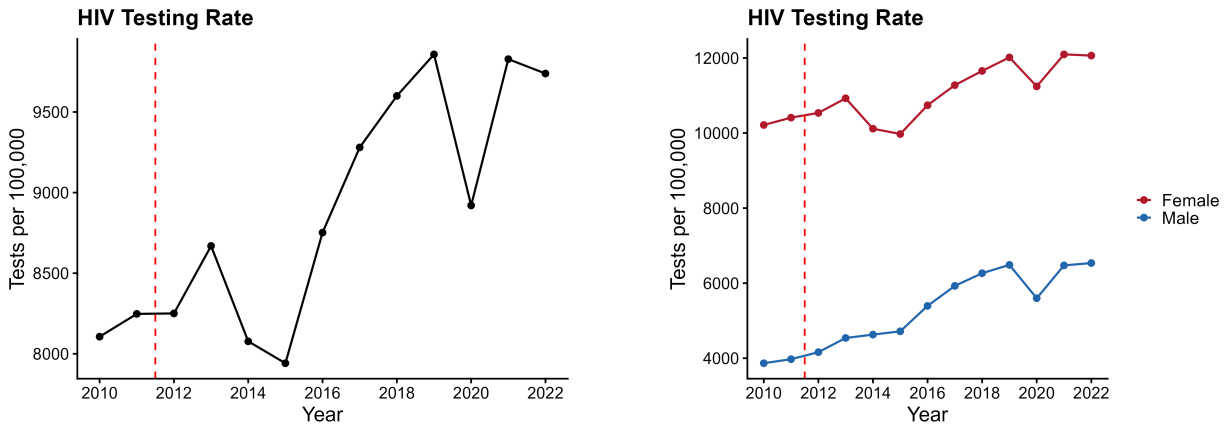
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A HIV Testing Check

Syphilis testing may be particularly difficult to identify in claims data because tests can be included in bundled codes or performed at free clinics and other settings that do not generate Medicaid claims. As a check, I examine the effect of MSS on HIV testing. People are supposed to be tested for HIV before initiating PrEP and periodically while taking it. Taking PrEP with undiagnosed HIV partially treats the virus and can lead to treatment-resistant infections, among other problems. If HIV testing increases while syphilis testing does not, that pattern would suggest that syphilis testing is not changing. If HIV testing decreases, this could reflect less concern about HIV and a lower frequency of testing. If estimates for both outcomes are imprecise, that highlights the general difficulty of identifying changes in this type of testing in claims data.

I construct the HIV testing rate as the number of HIV test events per 100,000 Medicaid

beneficiaries aged 15 and older in each state-year, using the CPT codes listed in Appendix B. Figure 25 shows that HIV testing among Medicaid beneficiaries aged 15 and older rose over the sample period, with a dip in 2020, and that testing rates are substantially higher among women than among men.



(a) Pooled

(b) By Sex

Figure 25: National HIV Testing Trends

The HIV testing rate is the number of HIV test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Rates are aggregated across all sample states in each year.

The event-study coefficients in Figures 26 and 27 show no clear post-2012 pattern. Table 13 shows that the estimated effect of MSS on the HIV testing rate is positive but imprecisely estimated. The preferred estimate of 3,688.7 implies roughly 324 additional HIV tests per 100,000 beneficiaries for a one-standard-deviation higher MSS, about 5% of the 2011 baseline mean of 6,151. The male and female estimates in Table 14 are likewise positive and imprecise. This suggests either that there is a systematic imprecision in identifying sexually transmitted infection testing that affects both the syphilis and HIV testing estimates, or that the effects on the testing outcomes are both truly noisy and insignificant.

Dependent Variable:	Testing Rate	
Model:	(1)	(2)
Variables		
MSS × Post	4,821.4	3,688.7
	(4,210.5)	(4,090.9)
Demographic Controls	No	Yes
2011 Baseline Mean	6,151.25	6,151.25
Fixed-effects		
State	Yes	Yes
Year	Yes	Yes
Fit statistics		
Observations	647	647

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 13: Effect of MSS × Post on the HIV Testing Rate

The HIV testing rate is the number of HIV test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic, the share of enrollees aged 0-64 who are adults aged 19-64, and the male share of enrollees aged 19-64. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

	Male		Female	
Dependent Variable:	Testing Rate			
Model:	(1)	(2)	(3)	(4)
Variables				
MSS × Post	3,506.1	2,228.9	6,816.8	4,861.6
	(2,288.9)	(2,160.1)	(5,618.2)	(5,391.3)
Demographic Controls	No	Yes	No	Yes
2011 Baseline Mean	2,286.10	2,286.10	8,283.64	8,283.64
Fixed-effects				
State	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	647	647	647	647

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 14: Effect of MSS × Post on the HIV Testing Rate, Male and Female

The HIV testing rate is the number of HIV test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Columns (1) and (2) restrict the sample to male beneficiaries and columns (3) and (4) to female beneficiaries. The male and female specifications are estimated separately. Specifications marked "Yes" in the Demographic Controls row include controls, from ACS 1-year PUMS, for the shares of each state's male or female Medicaid enrollees aged 0-64 who are White non-Hispanic, Black non-Hispanic, and Hispanic and the share who are adults aged 19-64, computed among enrollees of the same sex as the sample. They also include an indicator for Medicaid expansion. The unit of observation is the state-year, 2010-2022, excluding DC. All specifications include state and year fixed effects. Standard errors clustered at the state level are reported in parentheses.

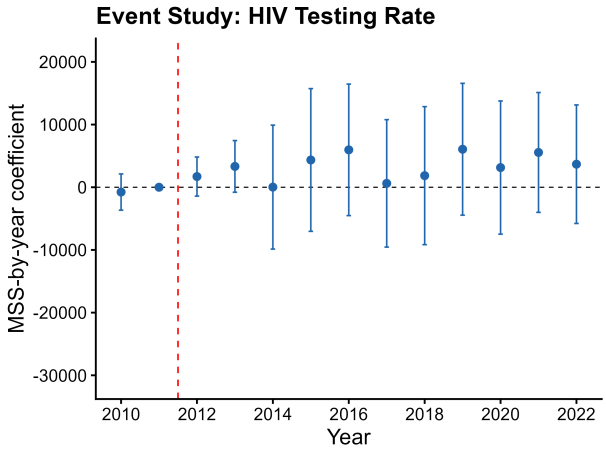
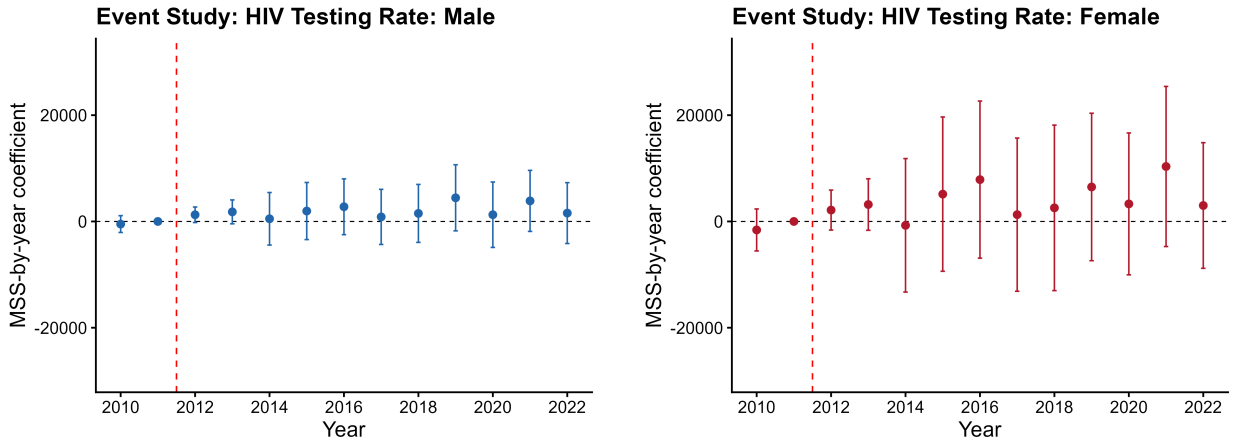


Figure 26: Event Study: HIV Testing Rate

The HIV testing rate is the number of HIV test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.



(a) Male

(b) Female

Figure 27: Event Studies: HIV Testing Rate, Male and Female

The HIV testing rate is the number of HIV test events per 100,000 Medicaid beneficiaries aged 15 and older in the state, measured annually. Male and female coefficients are estimated separately. Coefficients and 95% confidence intervals are from the event-study specification in Equation (3), with 2011 as the omitted reference year. Event-study estimates shown are from the specification with demographic controls.

B Code Definitions

Table 15: NDC Codes Used to Identify PrEP Drug Fills

NDC codes (11-digit, 5-4-2 format), by drug				
Truvada (emtricitabine/tenofovir disoproxil fumarate) 200/300 mg tablet (47 codes)				
00247-2104-01	00247-2104-07	00247-2104-21	00247-2104-30	24236-0303-01
24236-0303-02	24236-0303-09	35356-0070-03	35356-0070-06	35356-0070-30
50090-0870-00	50090-0870-02	50090-0870-03	50436-0701-01	51655-0325-24
51655-0325-52	51655-0325-53	51655-0325-99	52959-0969-03	53808-0805-01
54569-5588-00	54569-5588-02	54569-5588-03	54868-5141-00	55045-3481-03
55154-4425-00	55154-4425-04	55154-4425-06	58118-0100-08	61786-0596-02
61786-0596-38	61919-0669-02	61958-0701-01	61958-0702-01	61958-0702-02
63285-0020-30	63629-7581-01	63629-7581-02	63629-7581-03	66336-0032-03
67263-0260-30	68071-2112-03	68258-1983-03	69189-0701-01	70518-0097-00
70518-0097-01	70518-0097-02			
Generic emtricitabine/tenofovir disoproxil fumarate 200/300 mg tablet (102 codes)				
00093-7607-56	00093-7704-56	00378-1930-05	00378-1930-77	00378-1930-93
00904-7172-07	16714-0534-01	31722-0560-30	33342-0106-07	42291-0439-30
42291-0980-30	42385-0913-30	42385-0913-90	42385-0953-30	42385-0953-90
42543-0719-04	50090-5957-01	50090-6494-01	50090-7719-01	50090-8028-01
51407-0112-30	53002-1653-01	53002-1653-02	53002-1653-03	53002-1653-05
53002-1653-06	53002-1653-07	53002-1653-09	55154-0082-05	60219-2095-03
60505-4202-03	60687-0719-25	60687-0719-95	63629-8426-01	63629-8426-02
64380-0719-04	65862-0354-30	67296-1237-03	67296-1237-04	67296-1433-03
67296-1440-03	67296-1440-07	67296-1459-03	67296-1830-07	67296-1895-03
67296-1895-05	67296-1895-08	67296-2198-03	67296-2198-05	68071-3695-03
68071-3831-03	68071-3986-03	68180-0287-01	68180-0287-06	69097-0209-02
69097-0741-02	69238-1527-03	69238-2095-03	70518-2937-00	70518-2937-01
70518-3077-00	70518-3077-01	70518-3077-02	70518-3696-00	70518-3696-01
70518-3696-02	70710-1367-02	70710-1367-03	70710-1367-04	70710-1367-09
70771-1594-02	70771-1594-03	70771-1594-04	70771-1594-09	70771-1709-02
70771-1709-03	70771-1709-04	70771-1709-09	71205-0776-03	71205-0776-07
71205-0776-30	71205-0776-60	71205-0776-90	71335-1766-01	71335-1766-02
71335-1766-03	71335-1844-01	71335-1844-02	71335-2650-01	71335-2650-02
71335-2650-03	71335-2650-04	72189-0156-02	72189-0227-02	72189-0312-03
72189-0403-03	76282-0677-30	82009-0109-30	82804-0086-03	82804-0086-07
82982-0056-03	82982-0068-03			

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Table 15 – continued

NDC codes (11-digit, 5-4-2 format), by drug				
Descovy (emtricitabine/tenofovir alafenamide) 200/25 mg tablet (3 codes)				
61958-2002-01	61958-2002-02	70518-0230-00		
Apretude (cabotegravir extended-release injectable suspension) 600 mg/3 mL (3 codes)				
49702-0238-03	49702-0264-23	49702-0280-63		

Notes: Codes are all prescription National Drug Codes (NDCs) for products whose only active ingredients are emtricitabine and tenofovir disoproxil fumarate (Truvada and generics) or emtricitabine and tenofovir alafenamide (Descovy), and for injectable cabotegravir labeled as Apretude. Products were identified by active ingredient from the FDA National Drug Code Directory (U.S. Food and Drug Administration, 2026) and from historical NDC records in the National Library of Medicine's RxNorm, accessed through the RxNav API (National Library of Medicine, 2026; Nelson et al., 2011). Lower-strength tablets used only for pediatric HIV treatment (emtricitabine/tenofovir disoproxil fumarate 100/150, 133/200, and 167/250 mg, and emtricitabine/tenofovir alafenamide 120/15 mg) are excluded. FDA package codes are converted from their 10-digit formats (4-4-2, 5-3-2, or 5-4-1) to the 11-digit 5-4-2 format used in Medicaid claims by zero-padding each segment. A Truvada, generic Truvada, or Descovy fill is classified as PrEP only if the beneficiary has no claim for a co-prescribed antiretroviral (Table 16) within 90 days of the fill and the beneficiary's cumulative supply exceeds 30 days. Apretude is classified as PrEP regardless of other prescriptions and is also identified in outpatient claims by HCPCS code J0739.

Table 16: NDC Codes Used to Identify Co-Prescribed Antiretroviral Drugs

NDC codes (11-digit, 5-4-2 format), by drug				
Protease inhibitors				
Atazanavir (Reyataz) (126 codes)				
00003-3622-12	00003-3623-12	00003-3624-12	00003-3631-12	00003-3638-10
00093-5526-06	00093-5527-06	00093-5528-56	00179-1610-70	00179-1610-71
00179-1610-72	00179-1665-70	00179-1665-71	00179-1665-72	00247-2105-14
00247-2105-30	00247-2105-60	00247-2106-14	00247-2106-30	00247-2106-60
00904-6875-04	16714-0860-01	16714-0861-01	16714-0862-01	24236-0288-01
24236-0288-02	24236-0288-09	24236-0289-01	24236-0289-02	24236-0667-01
24236-0667-02	31722-0653-10	31722-0653-31	31722-0653-60	31722-0654-10
31722-0654-31	31722-0654-60	31722-0655-05	31722-0655-30	31722-0655-31
35356-0068-60	35356-0114-06	35356-0114-30	35356-0207-60	42385-0920-06
42385-0920-48	42385-0920-60	42385-0921-06	42385-0921-48	42385-0921-60

Continued on next page

Table 16 – continued

NDC codes (11-digit, 5-4-2 format), by drug				
42385-0922-06	42385-0922-30	42385-0922-37	50090-1007-00	50090-1581-00
51407-0171-60	51407-0172-60	51407-0173-30	51655-0302-52	51655-0302-53
51655-0302-87	51655-0302-99	54569-5530-00	54569-5532-00	54569-5864-00
54868-4854-00	54868-4857-00	54868-5838-00	54898-4854-00	59762-0408-06
59762-0409-06	59762-0410-03	60687-0399-25	60687-0399-95	65862-0710-03
65862-0710-05	65862-0710-10	65862-0710-60	65862-0711-03	65862-0711-05
65862-0711-10	65862-0711-18	65862-0711-60	65862-0712-03	65862-0712-05
65862-0712-10	65862-0712-18	65862-0712-60	65862-0713-03	65862-0713-05
65862-0713-10	65862-0713-30	65862-0713-90	68258-1984-06	68258-1985-03
68258-9142-01	69097-0443-02	69097-0443-03	69097-0444-02	69097-0444-03
69097-0445-02	69097-0445-03	69097-0446-02	69097-0446-03	69189-3631-01
69238-1135-06	69238-1136-06	69238-1137-06	69238-1138-03	70518-0469-00
70518-0725-00	70518-0946-00	70518-1365-00	70518-2832-00	70710-1050-01
70710-1050-06	70710-1051-01	70710-1051-06	70710-1052-01	70710-1052-03
70771-1591-01	70771-1591-06	70771-1592-01	70771-1592-06	70771-1593-01
70771-1593-03				
Atazanavir/cobicistat (Evotaz) (3 codes)				
00003-3641-11	50090-1724-00	54569-6595-00		
Darunavir (Prezista) (73 codes)				
00480-7736-06	24236-0299-01	24236-0299-02	24236-0927-02	31722-0088-60
31722-0089-30	31722-0567-60	31722-0568-60	35356-0113-30	35356-0284-60
43598-0704-05	43598-0704-60	43598-0705-05	43598-0705-30	50090-1327-00
50090-1491-00	52125-0613-02	53808-0773-01	54569-5814-00	54569-6086-00
54569-6366-00	54569-6403-00	54569-8140-00	54868-5631-00	54868-6369-00
59604-0560-01	59604-0562-01	59604-0563-01	59604-0564-01	59651-0085-60
59651-0086-30	59676-0560-01	59676-0561-01	59676-0562-01	59676-0563-01
59676-0564-01	59676-0565-01	59676-0566-30	60219-1144-06	60219-1145-03
60687-0808-11	60687-0808-21	60687-0819-11	60687-0819-21	63629-5686-01
67263-0590-60	68071-3910-03	68180-0345-07	68180-0346-06	68180-0346-09
68258-1980-06	69097-0244-02	69097-0248-03	69189-0566-01	70518-0689-00
70518-1483-00	70771-1745-05	70771-1745-06	70771-1745-08	70771-1746-03
70771-1746-05	70771-1746-09	71335-2611-01	72205-0184-60	72205-0185-30
72578-0147-05	72578-0147-14	72578-0147-28	72578-0148-05	72578-0148-06
72578-0148-16	76282-0737-60	76282-0738-30		

Continued on next page

Table 16 – continued

NDC codes (11-digit, 5-4-2 format), by drug				
Darunavir/cobicistat (Prezcobix) (6 codes)				
50090-1723-00	54569-6594-00	59676-0575-30	59676-0577-30	59676-0578-30
70518-0714-00				
Fosamprenavir (Lexiva) (24 codes)				
00173-0721-00	00173-0727-00	00378-3520-80	00378-3520-91	24236-0293-01
24236-0293-02	24236-0293-09	35356-0067-06	35356-0067-60	49349-0811-02
49702-0207-18	49702-0208-53	51947-8691-01	53808-0281-01	53808-1010-01
53873-0727-00	54569-5550-00	54868-4954-00	63304-0583-01	63304-0583-30
63304-0583-60	67263-0387-60	68258-1994-06	69189-0210-01	
Indinavir (Crixivan) (43 codes)				
00006-0570-62	00006-0571-42	00006-0571-43	00006-0573-01	00006-0573-18
00006-0573-40	00006-0573-42	00006-0573-54	00006-0573-62	00006-0574-65
00247-1543-06	00247-1543-18	00247-1543-36	00247-1543-42	00247-1543-59
00247-1543-84	00247-1546-06	00247-1546-18	00247-1546-42	00247-1546-66
12634-0524-92	12634-0524-94	12634-0524-96	16590-0064-18	16590-0064-30
16590-0064-60	16590-0064-90	24236-0659-02	51129-1651-01	52959-0507-12
52959-0507-18	52959-0507-24	52959-0507-30	54569-8620-00	54868-4113-00
55045-3548-01	55887-0230-30	55887-0230-60	58016-0699-00	58016-0699-30
58016-0699-60	58016-0699-90	68071-0149-18		
Lopinavir/ritonavir (Kaletra) (84 codes)				
00007-6799-22	00007-6799-30	00074-0522-06	00074-0522-60	00074-1575-21
00074-2605-21	00074-3008-60	00074-3956-46	00074-3959-77	00074-4014-12
00074-6799-22	00074-6799-30	00179-1723-70	00179-1723-71	00179-1723-72
00490-7028-00	00490-7028-01	00490-7028-02	00490-7028-03	00490-7028-04
00490-7028-05	00490-7028-06	00490-7028-07	00490-7028-08	00490-7028-09
00490-7028-10	00490-7028-30	00490-7028-60	00490-7028-90	00527-1947-48
11014-0019-01	11014-1295-01	11819-0342-20	12613-0522-06	12613-6799-06
17856-6799-02	21695-0362-12	24236-0676-01	24236-0676-02	24236-0676-06
24236-0676-13	31722-0556-12	31722-0556-31	31722-0556-32	31722-0556-60
31722-0603-12	31722-0603-60	33342-0163-09	33342-0164-09	35356-0111-60
35356-0112-01	35356-0112-30	42385-0933-60	42385-0934-12	49349-0024-02
49349-0024-06	49349-0024-13	49349-0024-21	51129-3613-01	51129-3613-02
52959-0134-18	52959-0968-12	53808-0276-01	54569-5525-00	54569-5752-00
54569-5752-01	54569-5752-02	54569-6199-00	54868-5566-00	55045-3482-01

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Table 16 – continued

NDC codes (11-digit, 5-4-2 format), by drug				
55154-6955-00	55154-6955-04	55154-6955-05	55154-6955-06	55289-0947-12
66336-0624-12	67263-0232-12	68071-0748-12	68071-2348-02	68071-2949-02
68258-1972-01	70518-0091-00	70518-0091-01	70518-0091-02	
Nelfinavir (Viracept) (56 codes)				
00179-1572-70	00179-1572-71	00179-1572-72	00179-1787-70	00179-1787-71
00179-1787-72	00247-1716-03	00247-1716-09	00247-1716-30	00247-1716-63
24236-0287-01	24236-0287-02	24236-0287-09	24236-0287-21	24236-0414-02
24236-0414-09	35356-0117-01	49999-0431-03	52959-0289-06	52959-0289-30
53808-0809-01	53869-0010-03	53869-0027-07	54569-4543-00	54569-4543-01
54569-4543-02	54569-4543-03	54569-4543-04	54569-4543-05	54569-5573-00
54868-3947-00	55045-2682-06	55045-2682-08	55045-2682-09	55154-6425-00
55154-6425-04	55154-6425-06	55289-0477-27	55370-0560-11	55370-0560-27
55370-0701-44	58016-0455-00	58016-0455-02	58016-0455-30	58016-0455-60
58016-0455-90	60760-0010-18	60760-0010-63	63010-0010-27	63010-0010-30
63010-0011-90	63010-0027-70	63285-0010-27	66267-0514-18	66267-0514-63
71114-4206-07				
Ritonavir (Norvir) (59 codes)				
00054-0407-13	00054-0407-23	00069-1345-01	00069-1345-05	00069-1345-11
00069-1735-05	00069-1735-11	00069-3085-01	00069-3085-11	00074-1940-63
00074-2340-30	00074-3333-30	00074-3399-30	00074-6633-15	00074-6633-22
00074-6633-30	00074-9492-02	00074-9492-54	00247-2194-07	11014-0032-01
11014-1233-01	31722-0597-12	31722-0597-30	35356-0138-30	50090-1162-00
53808-1119-01	54569-4613-00	54569-4792-00	54569-5656-00	54569-6170-00
54569-6170-01	54868-3782-00	54868-3782-01	54868-3782-02	54868-3782-03
60687-0364-25	60687-0364-95	60687-0420-25	60687-0420-95	61786-0386-02
63629-5702-01	65162-0061-06	65162-0061-32	65862-0687-01	65862-0687-05
65862-0687-30	65862-0687-99	67046-1251-03	67296-1234-04	68071-3898-03
68258-1987-03	68382-0696-06	69097-0655-02	69097-0655-08	69189-3335-01
70518-1086-00	70518-1519-00	70518-2007-00	71335-2610-01	
Saquinavir (Invirase) (22 codes)				
00004-0244-51	00004-0244-99	00004-0245-15	00004-0245-99	00004-0246-48
12806-0245-09	12806-5530-09	24236-0666-02	54569-4242-01	54569-4563-00
54569-4563-01	54569-5664-00	54868-3699-00	54868-3699-01	54868-3699-02
54868-4110-00	55361-1169-09	60346-1018-02	60346-1018-09	60491-0336-90
66311-0244-01	66311-0244-09			

Continued on next page

Table 16 – continued

NDC codes (11-digit, 5-4-2 format), by drug				
Tipranavir (Aptivus) (7 codes)				
00597-0002-01	00597-0003-01	00597-0003-02	11014-0052-01	11014-1307-01
14288-0951-01	54569-5719-00			
Non-nucleoside reverse transcriptase inhibitors (NNRTIs)				
Delavirdine (Rescriptor) (11 codes)				
00009-7576-01	00009-7576-18	00179-1592-70	00179-1592-71	00179-1592-72
54569-4562-00	54868-4520-00	63010-0020-36	63010-0021-18	65427-0757-18
65427-0761-36				
Doravirine (Pifeltro) (5 codes)				
00006-3069-01	00006-5092-01	00006-5092-59	50090-6237-00	50090-6268-00
Efavirenz (Sustiva) (43 codes)				
00056-0470-30	00056-0473-30	00056-0474-92	00056-0510-30	00056-0510-75
00378-2233-93	12634-0758-97	24236-0282-01	24236-0282-02	24236-0283-02
31722-0504-25	31722-0504-30	33342-0013-07	35356-0069-90	35356-0115-06
35356-0115-30	42543-0889-04	51407-0382-30	54569-4611-00	54569-5374-00
54868-4668-00	64380-0705-01	64380-0889-04	64980-0406-03	64980-0407-09
65862-0049-10	65862-0049-30	65862-0104-10	65862-0104-30	65862-0105-10
65862-0105-30	65862-0106-09	65862-0106-10	65862-0106-30	65862-0106-90
65862-0774-30	69097-0301-02	70518-1013-00	70518-1100-00	70518-2962-00
70518-3258-00	72865-0172-25	72865-0172-30		
Etravirine (Intelence) (28 codes)				
10147-0570-01	10147-0571-01	24236-0679-02	24236-0679-21	42291-0473-12
42291-0474-60	53808-0787-01	54569-6102-00	54569-6230-00	54868-5864-00
54868-6368-00	59676-0570-01	59676-0571-01	59676-0572-01	60219-1720-07
60219-1721-07	60219-1722-06	62579-0570-01	68258-1982-06	69238-1720-07
69238-1721-07	69238-1722-06	69315-0285-20	69315-0286-06	69452-0254-22
69452-0255-17	80005-0112-12	80005-0113-09		
Nevirapine (Viramune) (91 codes)				
00054-0450-58	00054-0459-21	00054-3905-58	00054-4647-21	00054-4647-25
00054-8647-25	00179-1644-70	00179-1644-71	00179-1644-72	00378-4050-05
00378-4050-80	00378-4050-91	00378-4890-37	00378-4890-77	00378-4890-93
00378-6950-05	00378-6950-77	00378-6950-93	00597-0046-01	00597-0046-41
00597-0046-46	00597-0046-60	00597-0046-61	00597-0047-24	00781-5893-05

Continued on next page

Table 16 – continued

NDC codes (11-digit, 5-4-2 format), by drug				
00781-5893-31	13925-0500-60	24236-0291-01	24236-0291-02	24236-0291-09
31722-0505-01	31722-0505-05	31722-0505-10	31722-0505-60	33342-0004-09
33342-0004-24	33342-0004-25	33342-0238-07	33342-0238-12	35356-0071-06
35356-0071-60	35356-0071-66	35356-0072-24	42571-0131-29	42571-0131-60
47781-0100-60	47781-0317-30	50228-0307-05	50228-0307-30	50228-0332-10
50228-0332-30	51129-1722-01	51991-0331-06	53104-0166-02	53104-0166-06
53150-0209-06	53808-0680-01	53808-0808-01	53808-1069-01	54569-4561-00
54569-4561-01	54868-3844-00	54868-3844-01	55289-0392-03	55289-0392-63
60429-0298-60	60505-3788-00	60505-3788-03	60505-3788-06	60505-4058-00
60505-4058-03	60505-4058-05	61442-0470-60	64380-0709-03	65162-0209-06
65862-0027-10	65862-0027-60	65862-0057-11	65862-0057-24	65862-0932-05
65862-0932-90	65862-0932-99	65862-0933-05	65862-0933-30	65862-0933-90
65862-0933-99	67263-0434-60	69097-0233-44	69097-0233-46	69097-0403-02
69097-0403-11				
Rilpivirine (Edurant) (10 codes)				
42385-0999-30	42385-0999-31	49702-0243-02	49702-0243-61	49702-0249-02
59676-0278-01	59676-0280-01	59676-0280-90	70069-0848-30	96796-0278-01
Integrase inhibitors				
Dolutegravir (Tivicay) (16 codes)				
49702-0226-13	49702-0227-13	49702-0228-13	49702-0255-37	50090-1355-00
50090-1355-01	53002-1810-06	53808-1130-01	54569-6419-00	54569-6419-01
60760-0748-07	61786-0771-02	67296-1818-03	67296-1818-05	67296-1818-07
70518-1487-00				
Raltegravir (Isentress) (49 codes)				
00006-0227-61	00006-0473-61	00006-0477-61	00006-3080-01	00006-3603-01
00006-3603-60	00006-3603-61	24236-0665-02	31722-0584-60	35356-0110-06
35356-0110-60	49349-0430-02	49349-0430-10	49349-0430-12	49349-0430-21
50090-1085-00	50090-1085-01	50090-1085-02	50473-0014-00	53002-1686-01
53002-1686-04	53002-1686-05	53002-1686-06	53808-0650-01	54569-6034-00
54569-6034-01	54868-0117-00	61786-0843-02	61786-0843-10	61786-0843-12
61919-0706-02	61919-0706-04	61919-0706-06	66336-0214-06	67296-1235-07
68071-2113-06	68258-1981-06	69189-0227-01	70518-1621-00	70518-1621-01
70518-1621-02	70748-0327-07	71205-0777-06	71205-0777-14	71205-0777-30
71205-0777-60	71205-0777-90	76519-1130-06	82982-0055-06	

Continued on next page

Table 16 – continued

NDC codes (11-digit, 5-4-2 format), by drug				
Entry, attachment, and capsid inhibitors				
Enfuvirtide (Fuzeon) (9 codes)				
00004-0380-39	00004-0381-01	00004-0381-40	00009-0380-39	12806-0380-08
12806-0380-09	12871-0380-09	35356-0206-60	54569-5781-00	
Fostemsavir (Rukobia) (1 code)				
49702-0250-18				
Ibalizumab (Trogarzo) (2 codes)				
62064-0122-01	62064-0122-02			
Lenacapavir (Sunlenca) (6 codes)				
61958-3001-01	61958-3001-02	61958-3001-03	61958-3002-01	61958-3004-01
61958-3005-01				
Maraviroc (Selzentry) (40 codes)				
00069-0807-60	00069-0808-60	31722-0579-60	31722-0580-60	35356-0208-60
35356-0209-60	42858-0246-06	42858-0247-06	49702-0215-18	49702-0216-18
49702-0223-18	49702-0224-18	49702-0233-08	49702-0235-08	49702-0237-48
49702-0237-55	49702-0260-55	50090-1486-00	50090-6238-00	50090-8005-00
50090-8010-00	50436-0224-01	52125-0011-02	53869-0807-03	53869-0808-04
54569-6143-00	54569-6214-00	54868-5809-00	68475-0007-01	68475-0008-01
68554-3079-00	68554-3080-00	70710-1930-06	70710-1931-06	72319-0024-02
72319-0025-02	72865-0231-60	72865-0232-60	73141-0024-02	73141-0025-02
Pharmacokinetic boosters				
Cobicistat (Tybost) (1 code)				
61958-1401-01				

Notes: Codes are all prescription National Drug Codes (NDCs) for products whose active ingredients include an antiretroviral that is co-prescribed with a PrEP drug (Truvada or Descovy) in HIV treatment. These include protease inhibitors, non-nucleoside reverse transcriptase inhibitors, integrase inhibitors, entry, attachment, and capsid inhibitors, and the pharmacokinetic booster cobicistat. Products were identified by active ingredient from the FDA National Drug Code Directory (U.S. Food and Drug Administration, 2026) and from historical NDC records in the National Library of Medicine's RxNorm, accessed through the RxNav API (National Library of Medicine, 2026; Nelson et al., 2011), which retain NDCs for products discontinued before the current FDA listing. Other antiretrovirals used in HIV treatment are excluded because they are not taken together with

Truvada or Descovy. FDA package codes are converted from their 10-digit formats (4-4-2, 5-3-2, or 5-4-1) to the 11-digit 5-4-2 format used in Medicaid claims by zero-padding each segment. A Truvada, generic Truvada, or Descovy fill is classified as HIV treatment rather than PrEP if the same beneficiary has a claim for any of these codes within 90 days of the fill.

Table 17: ICD-9-CM and ICD-10-CM Diagnosis Codes for Syphilis

Syphilis Category	ICD-9-CM Code(s)	ICD-10-CM Code(s)
Congenital	090.0, 090.1, 090.2, 090.3, 090.4, 090.5, 090.6, 090.7, 090.9	A50.01-A50.09, A50.1, A50.2, A50.30-A50.39, A50.40-A50.49, A50.51-A50.59, A50.6, A50.7, A50.9
Early, symptomatic	091.0, 091.1, 091.2, 091.3, 091.4, 091.50, 091.51, 091.52, 091.61, 091.62, 091.69, 091.7, 091.81, 091.82, 091.89, 091.9	A51.0, A51.1, A51.2, A51.31, A51.32, A51.39, A51.41, A51.42, A51.43, A51.44, A51.45, A51.46, A51.49
Early, latent	092.0, 092.9	A51.5
Cardiovascular	093.0, 093.1, 093.20, 093.21, 093.22, 093.23, 093.24, 093.81, 093.82, 093.89, 093.9	A52.00-A52.09
Neurosyphilis	094.0, 094.1, 094.2, 094.3, 094.81, 094.82, 094.83, 094.84, 094.85, 094.86, 094.87, 094.89, 094.9	A52.10-A52.19, A52.2, A52.3
Other late, symptomatic	095.0, 095.1, 095.2, 095.3, 095.4, 095.5, 095.6, 095.7, 095.8, 095.9	A52.71-A52.79
Late, latent	096	A52.8
Unspecified / other	097.0, 097.1, 097.9	A52.9, A53.0, A53.9

Notes: Diagnosis codes follow the ICD-9-CM and ICD-10-CM classifications (National Center for Health Statistics, 2026a, 2026b). A syphilis case is defined using diagnosis codes from all categories listed in this table.

Table 18: CPT Codes for Syphilis Testing

Test Type	CPT Code	Description
Nontreponemal, qualitative	86592	RPR/VDRL/ART, qualitative
Nontreponemal, quantitative	86593	RPR/VDRL, quantitative (titer)
Treponemal	86780	Antibody to *T. pallidum* (e.g., TP-PA, EIA/CIA)

Notes: Syphilis tests are identified using the CPT codes for nontreponemal and treponemal antibody tests used by Lanier et al. (2023).

Table 19: CPT Codes for HIV Testing

Test Type	Code	Description
Antibody	86701	Antibody, HIV-1
Antibody	86702	Antibody, HIV-2
Antibody	86703	Antibody, HIV-1 and HIV-2, single result
Antigen/antibody combination	87389	HIV-1 antigen(s) with HIV-1 and HIV-2 antibodies, single result
Antigen	87390	HIV-1 antigen
Antigen	87391	HIV-2 antigen
Rapid antigen/antibody	87806	HIV-1 antigen(s) with HIV-1 and HIV-2 antibodies, rapid immunoassay with direct optical observation
Nucleic acid (HIV-1)	87534	HIV-1, direct probe technique
Nucleic acid (HIV-1)	87535	HIV-1, amplified probe technique
Nucleic acid (HIV-2)	87537	HIV-2, direct probe technique
Nucleic acid (HIV-2)	87538	HIV-2, amplified probe technique